

Apache Web Server

- Apache web server:
- The Apache Web server is a free HTTP (Web) server developed by the Apache Server Project.
- It is a reliable, efficient , and easily extensible Web Server
- Free open source code

Apache Web Server

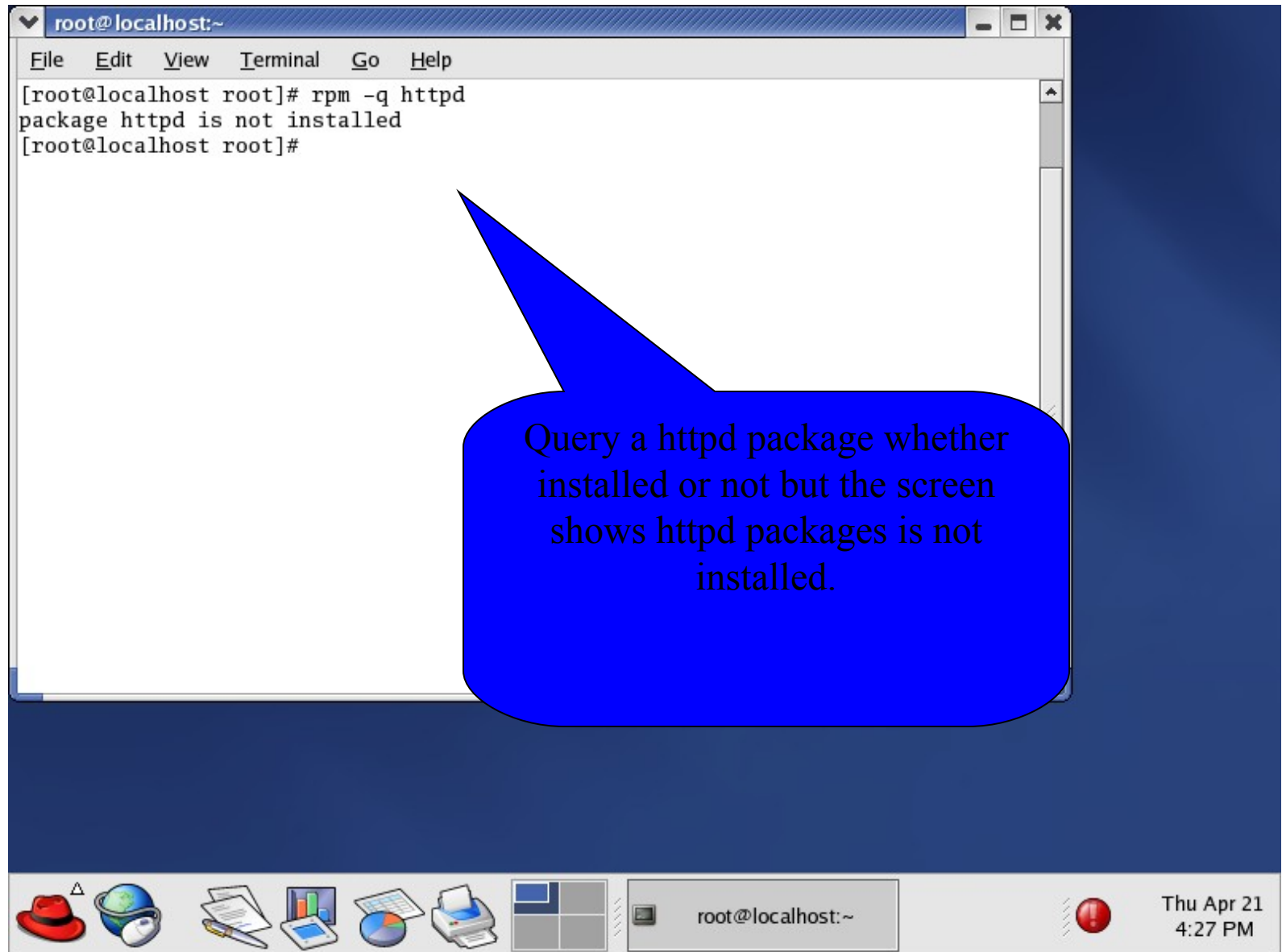
Important directories and configuration files

Web Site Directories	Description
/var/www	Directory for Apache Web site files on your Red Hat.
/var/www/html	Web site Web files.
/var/www/cgi-bin	CGI program files.
/var/www/usage	Apache Web server manual.
Configuration Files	
.htaccess	Directory-based configuration files. An .htaccess file holds directives to control access to files within the directory in which it is located.
/etc/httpd/conf	Directory for Apache Web server configuration files.
/etc/httpd/conf/httpd.conf	Primary Apache Web server configuration file (from version 1.3.4, it holds all configuration directives).

Service profile

- Packages - httpd
- Configurations file -
/etc/httpd/conf/httpd.conf
- Daemons - httpd
- Portnumber - 80

Web server configuration



Web server configuration

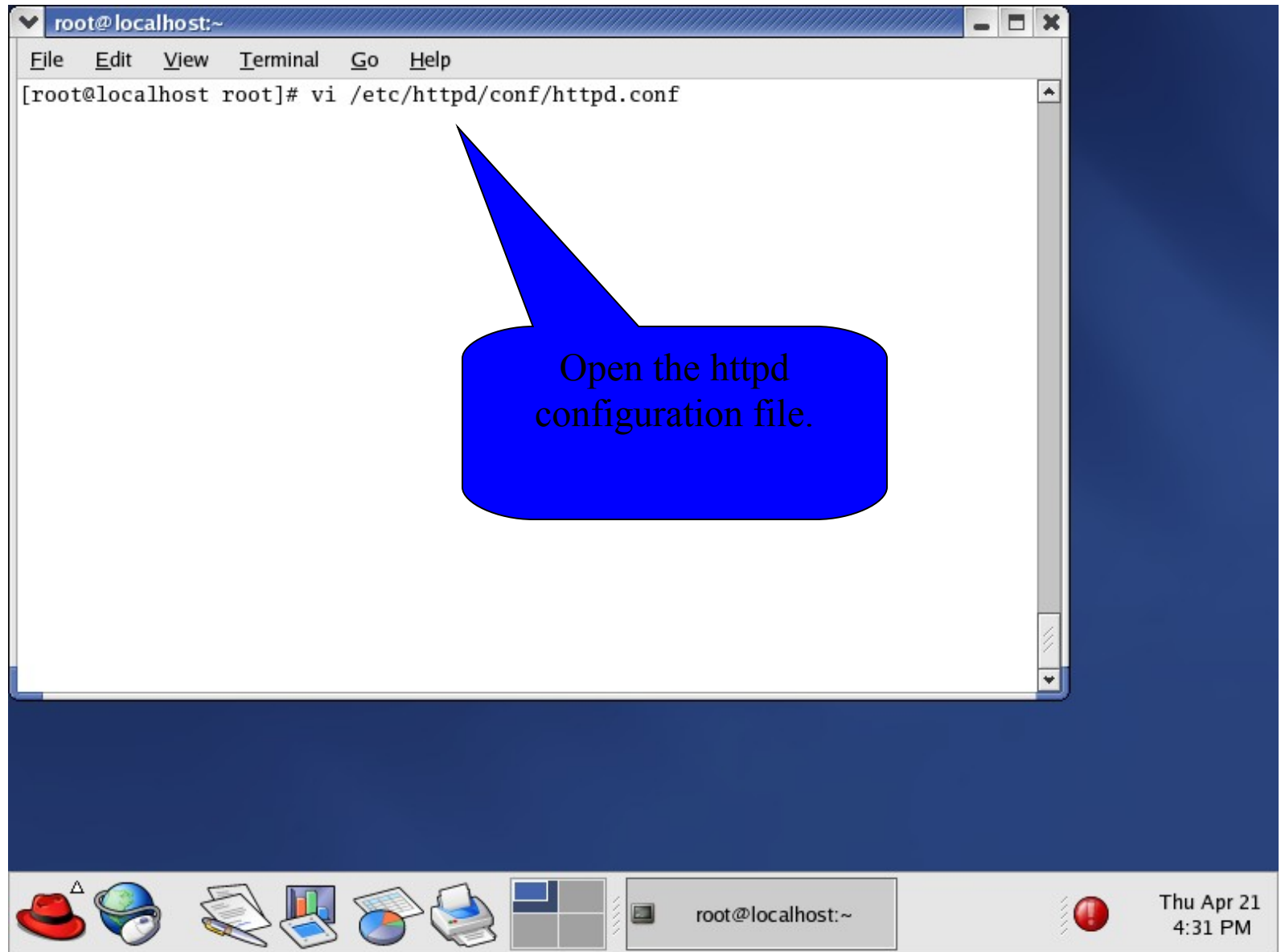
```
root@localhost:~  
File Edit View Terminal Go Help  
[root@localhost root]# cd /mnt/cdrom/RedHat/RPMS/  
[root@localhost RPMS]# rpm -ivh httpd-2.0.46-25.ent.i386.rpm  
warning: httpd-2.0.46-25.ent.i386.rpm: V3 DSA signature: NOKEY, key ID db42a60e  
Preparing... ##### [100%]  
1:httpd ##### [100%]  
[root@localhost RPMS]# cd  
[root@localhost root]# rpm -q httpd  
httpd-2.0.46-25.ent  
[root@localhost root]#
```

Now query the package
it show package is
installed.

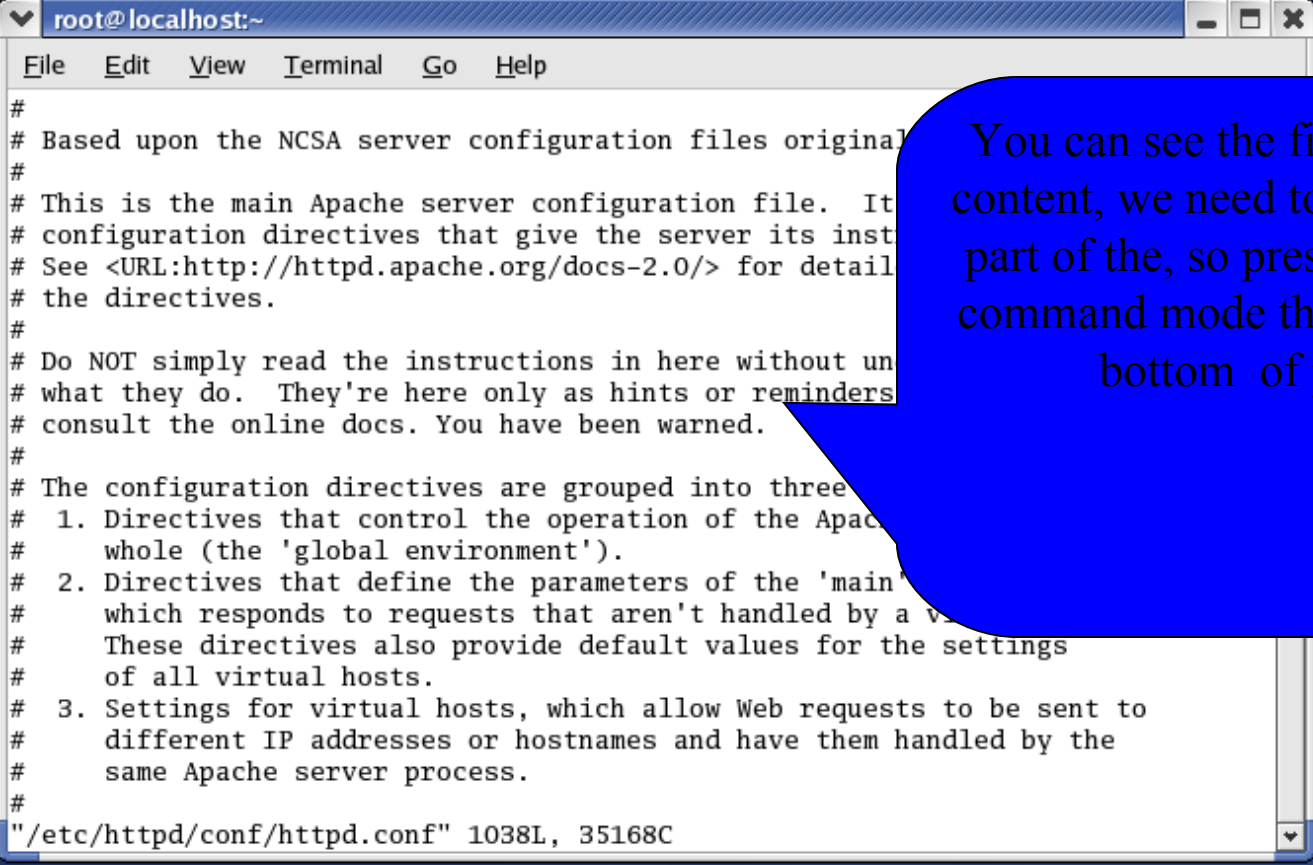


Sat Apr 23
1:06 PM

Web server configuration



Web server configuration



root@localhost:~

File Edit View Terminal Go Help

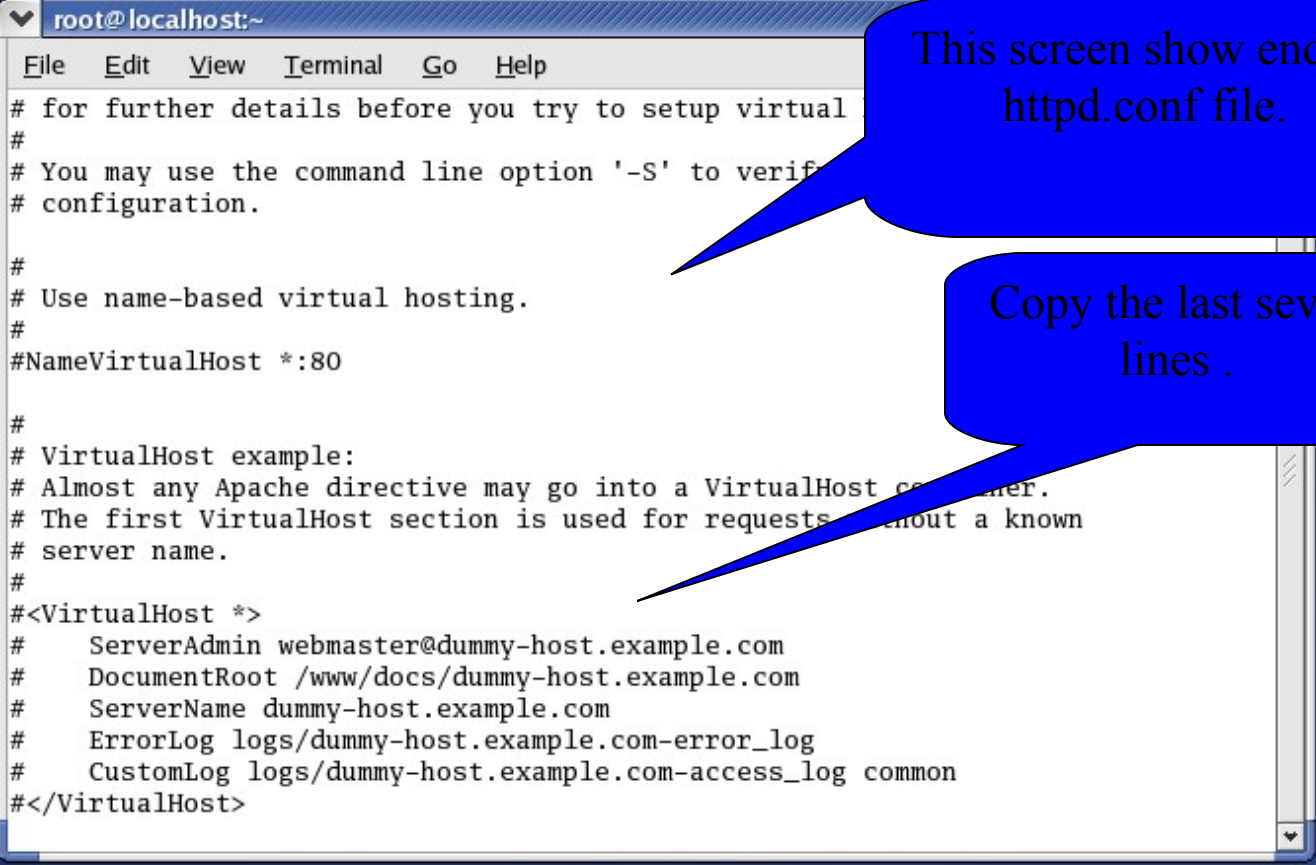
```
#  
# Based upon the NCSA server configuration files original  
#  
# This is the main Apache server configuration file. It  
# configuration directives that give the server its inst  
# See <URL:http://httpd.apache.org/docs-2.0/> for detail  
# the directives.  
#  
# Do NOT simply read the instructions in here without un  
# what they do. They're here only as hints or reminders  
# consult the online docs. You have been warned.  
#  
# The configuration directives are grouped into three  
# 1. Directives that control the operation of the Apac  
# whole (the 'global environment').  
# 2. Directives that define the parameters of the 'main'  
# which responds to requests that aren't handled by a v  
# These directives also provide default values for the settings  
# of all virtual hosts.  
# 3. Settings for virtual hosts, which allow Web requests to be sent to  
# different IP addresses or hostnames and have them handled by the  
# same Apache server process.  
#  
"/etc/httpd/conf/httpd.conf" 1038L, 35168C
```

You can see the first page file content, we need to edit the end part of the, so press shift+G in command mode the cursor will bottom of page.

root@localhost:~

Thu Apr 21 4:31 PM

Web server configuration



root@localhost:~

File Edit View Terminal Go Help

```
# for further details before you try to setup virtual
#
# You may use the command line option '-S' to verify
# configuration.
#
# Use name-based virtual hosting.
#
#NameVirtualHost *:80
#
# VirtualHost example:
# Almost any Apache directive may go into a VirtualHost container.
# The first VirtualHost section is used for requests without a known
# server name.
#
#<VirtualHost *>
#   ServerAdmin webmaster@dummy-host.example.com
#   DocumentRoot /www/docs/dummy-host.example.com
#   ServerName dummy-host.example.com
#   ErrorLog logs/dummy-host.example.com-error_log
#   CustomLog logs/dummy-host.example.com-access_log common
#</VirtualHost>
```

This screen show end of httpd.conf file.

Copy the last seven lines .

root@localhost:~

Thu Apr 21 4:31 PM

Web server configuration

root@localhost:~

File Edit View Terminal Go Help

```
#NameVirtualHost *:80

#
# VirtualHost example:
# Almost any Apache directive may go into a VirtualHost section.
# The first VirtualHost section is used for requests with no
# server name.
#
#<VirtualHost *>
#   ServerAdmin webmaster@dummy-host.example.com
#   DocumentRoot /www/docs/dummy-host.example.com
#   ServerName dummy-host.example.com
#   ErrorLog logs/dummy-host.example.com-error_log
#   CustomLog logs/dummy-host.example.com-access_log common
#</VirtualHost>

<VirtualHost 192.168.0.20>
    ServerAdmin root@station20.example.com
    DocumentRoot /var/www/html
    ServerName station20.example.com
    ErrorLog logs/station20.example.com-error_log
    CustomLog logs/station20.example.com-access_log common
</VirtualHost>
-- INSERT --
```

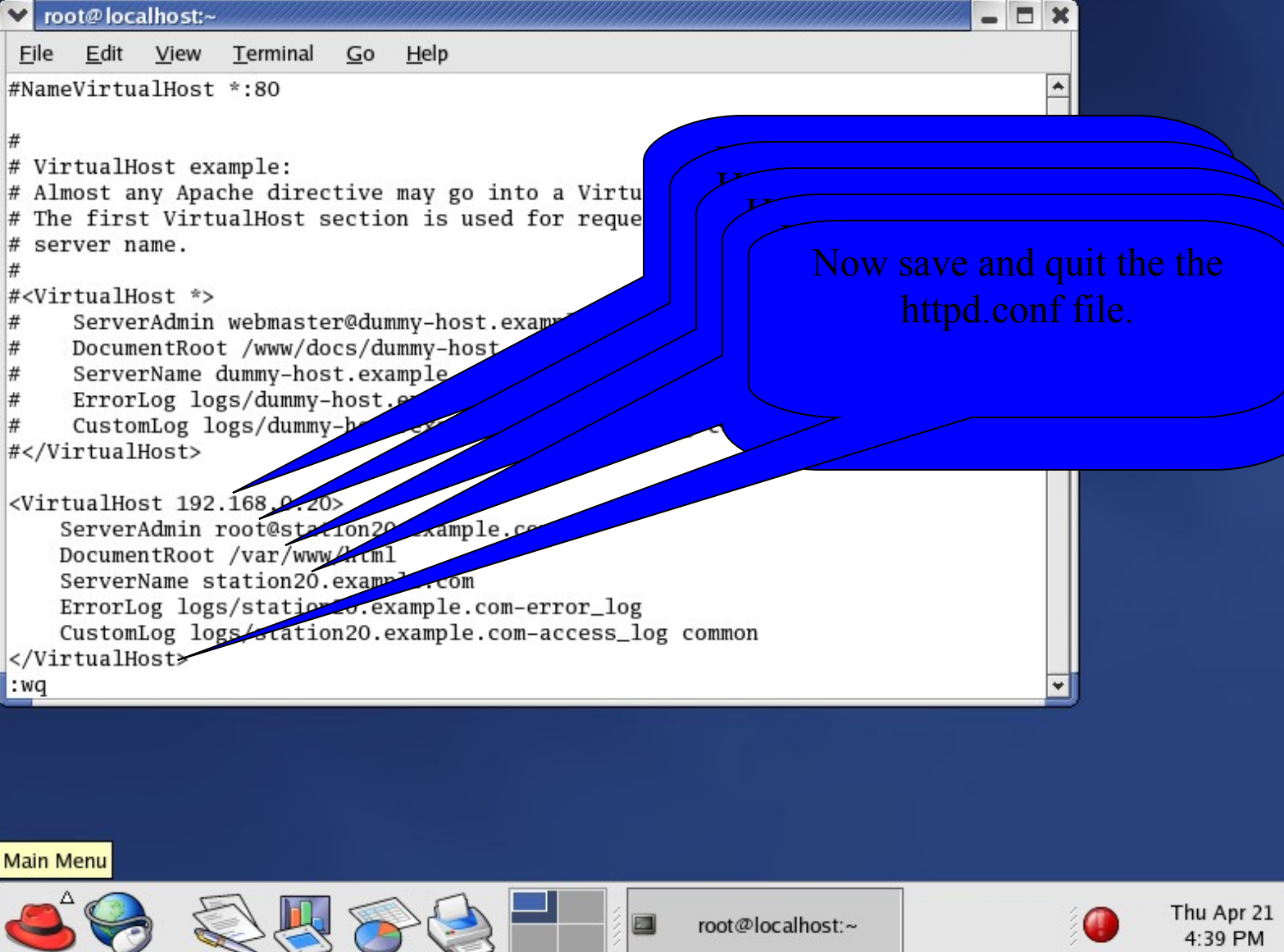
Paste the copied lines here .

Main Menu

root@localhost:~

Thu Apr 21 4:35 PM

Web server configuration



The image shows a terminal window titled 'root@localhost:~' with a menu bar (File, Edit, View, Terminal, Go, Help). The terminal displays the configuration for the `VirtualHost` directive in `httpd.conf`. The configuration includes a commented-out example and an active configuration for `192.168.0.20`. A blue callout bubble with a black border points to the configuration lines, containing the text: 'Now save and quit the the httpd.conf file.'

```
#NameVirtualHost *:80

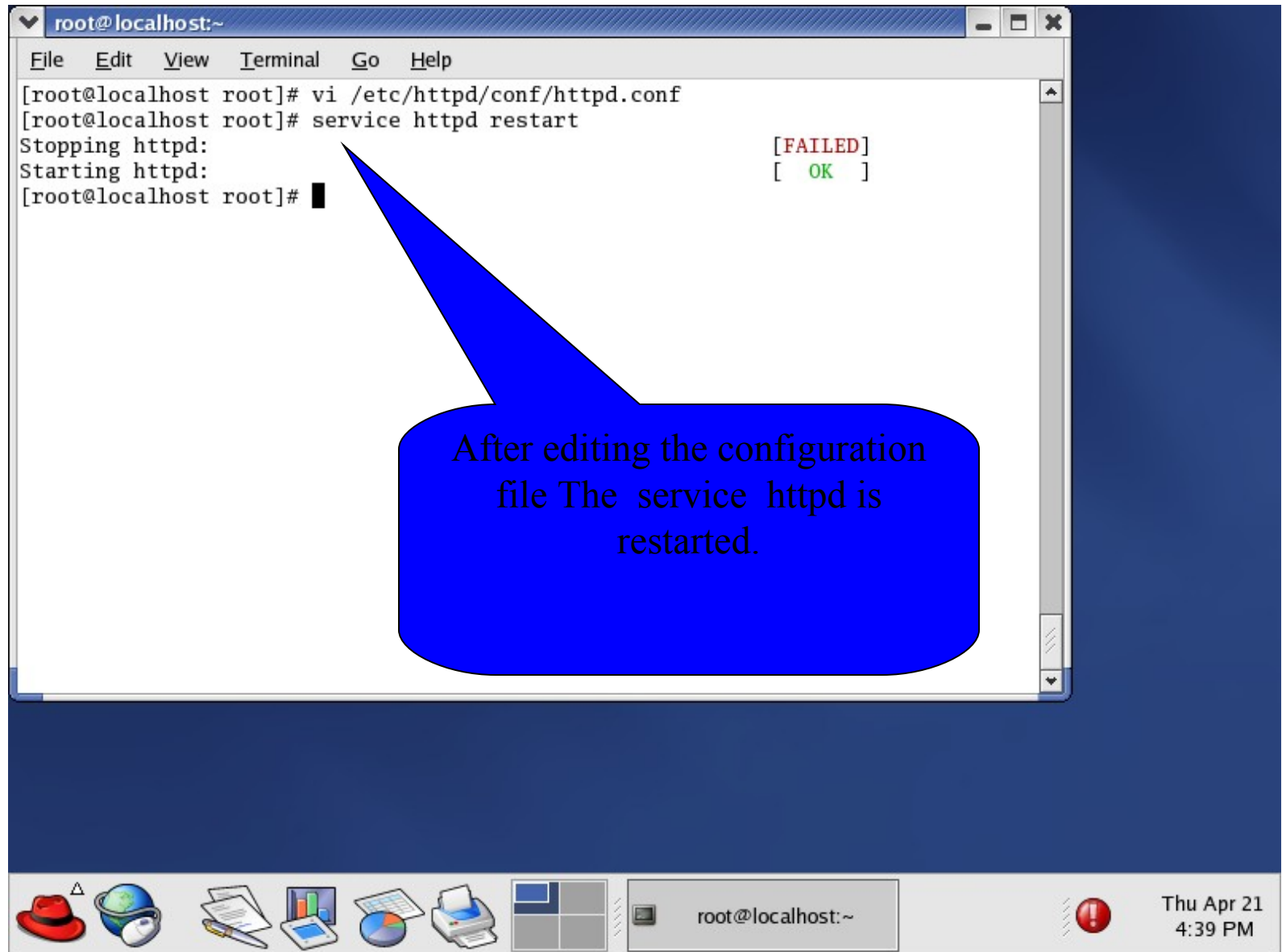
#
# VirtualHost example:
# Almost any Apache directive may go into a VirtualHost section.
# The first VirtualHost section is used for requests without a specific
# server name.
#
#<VirtualHost *>
#   ServerAdmin webmaster@dummy-host.example.com
#   DocumentRoot /www/docs/dummy-host.example.com
#   ServerName dummy-host.example.com
#   ErrorLog logs/dummy-host.example.com-error_log
#   CustomLog logs/dummy-host.example.com-access_log common
#</VirtualHost>

<VirtualHost 192.168.0.20>
    ServerAdmin root@station20.example.com
    DocumentRoot /var/www/html
    ServerName station20.example.com
    ErrorLog logs/station20.example.com-error_log
    CustomLog logs/station20.example.com-access_log common
</VirtualHost>
:wq
```

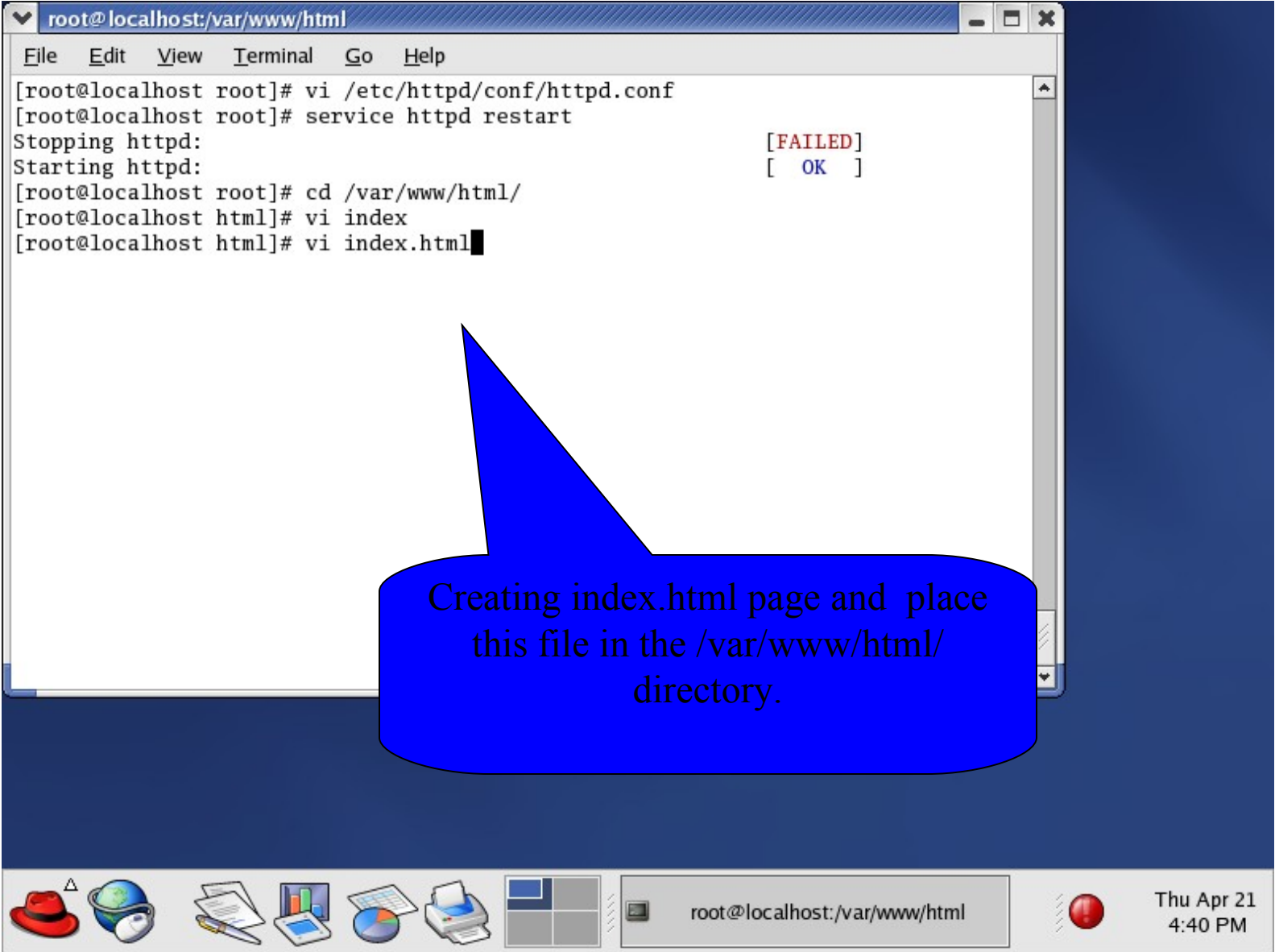
Main Menu

root@localhost:~ Thu Apr 21 4:39 PM

Web server configuration



Web server configuration



The image shows a terminal window with a blue title bar and a menu bar (File, Edit, View, Terminal, Go, Help). The terminal text is as follows:

```
root@localhost: /var/www/html
[root@localhost root]# vi /etc/httpd/conf/httpd.conf
[root@localhost root]# service httpd restart
Stopping httpd: [FAILED]
Starting httpd: [ OK ]
[root@localhost root]# cd /var/www/html/
[root@localhost html]# vi index
[root@localhost html]# vi index.html
```

A blue callout box with a pointer to the terminal contains the text:

Creating index.html page and place this file in the /var/www/html/ directory.

The desktop environment at the bottom includes a taskbar with icons for a red hat, a globe, a notepad, a bar chart, a pie chart, a printer, and a window manager. The system tray shows the terminal path, a red indicator light, and the date/time: Thu Apr 21 4:40 PM.

```

root@localhost:~# cat > station20.html
<html>
<title>station20.example.com</title>
<body>
<marquee>Welcome to the test page to learn webserver </marquee>
</body>
</html>
:~#
:wq

```

After typing save and quit the file.

OpenOffice.org Writer
Word processor

root@localhost:~# Thu Apr 21 4:44 PM

```

root@localhost:~# cat > station20.html
<html>
<title>station20.example.com</title>
<body>
<marquee>Welcome to the test page to learn webserver </marquee>
</body>
</html>
:~#
:wq

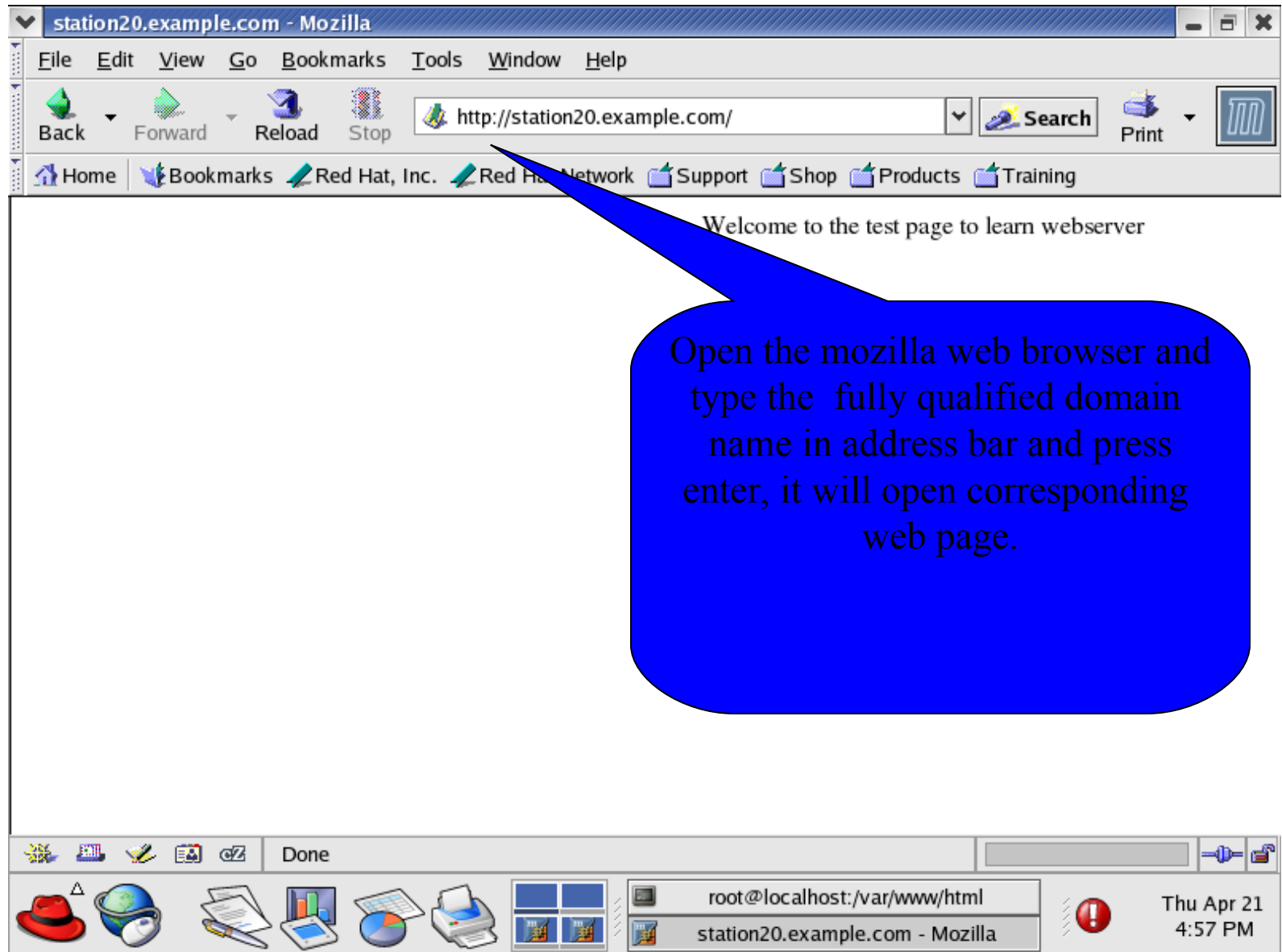
```

After typing save and quit the file.

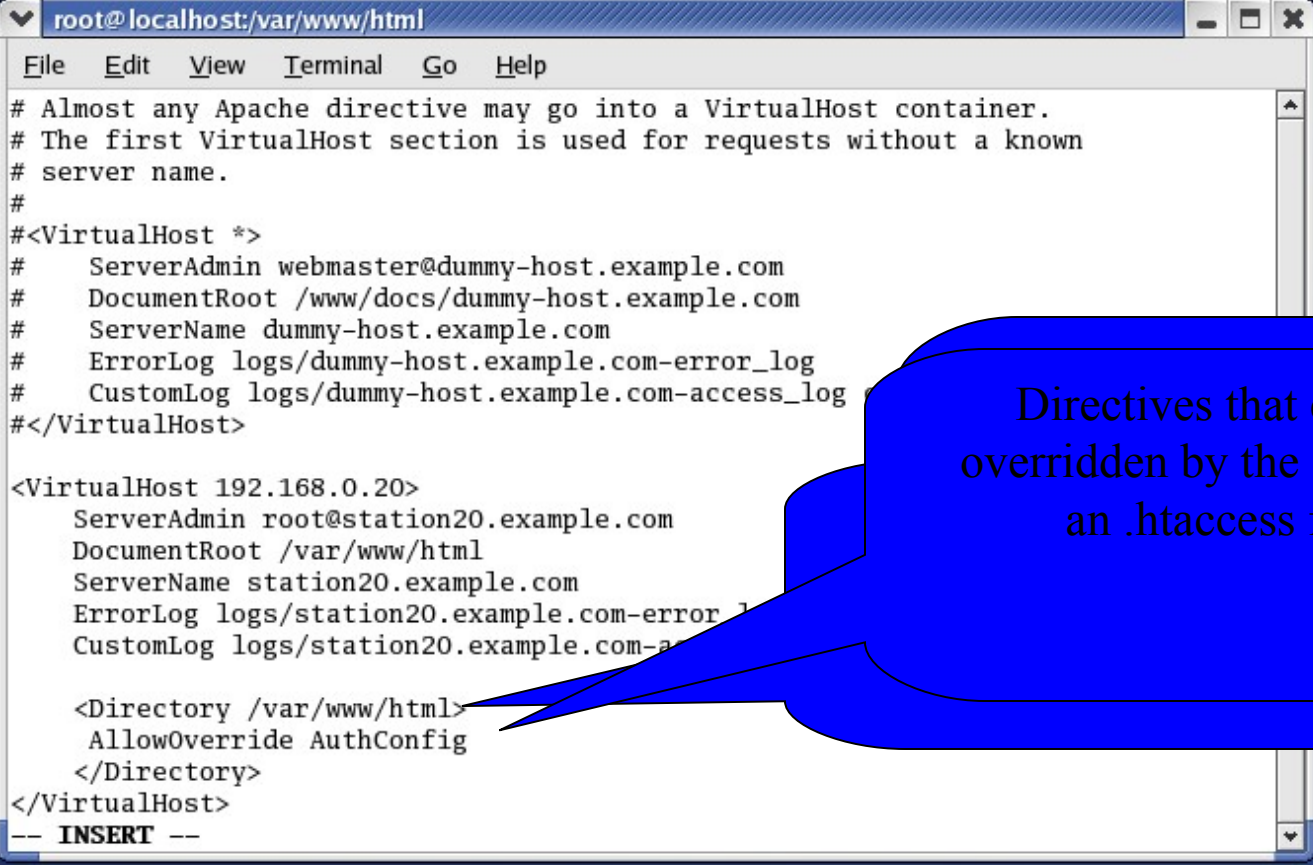
OpenOffice.org Writer
Word processor

root@localhost:~# Thu Apr 21 4:44 PM

Web server configuration



Web server configuration



```
root@localhost:/var/www/html
File Edit View Terminal Go Help
# Almost any Apache directive may go into a VirtualHost container.
# The first VirtualHost section is used for requests without a known
# server name.
#
#<VirtualHost *>
#   ServerAdmin webmaster@dummy-host.example.com
#   DocumentRoot /www/docs/dummy-host.example.com
#   ServerName dummy-host.example.com
#   ErrorLog logs/dummy-host.example.com-error_log
#   CustomLog logs/dummy-host.example.com-access_log
#</VirtualHost>

<VirtualHost 192.168.0.20>
  ServerAdmin root@station20.example.com
  DocumentRoot /var/www/html
  ServerName station20.example.com
  ErrorLog logs/station20.example.com-error_log
  CustomLog logs/station20.example.com-access_log

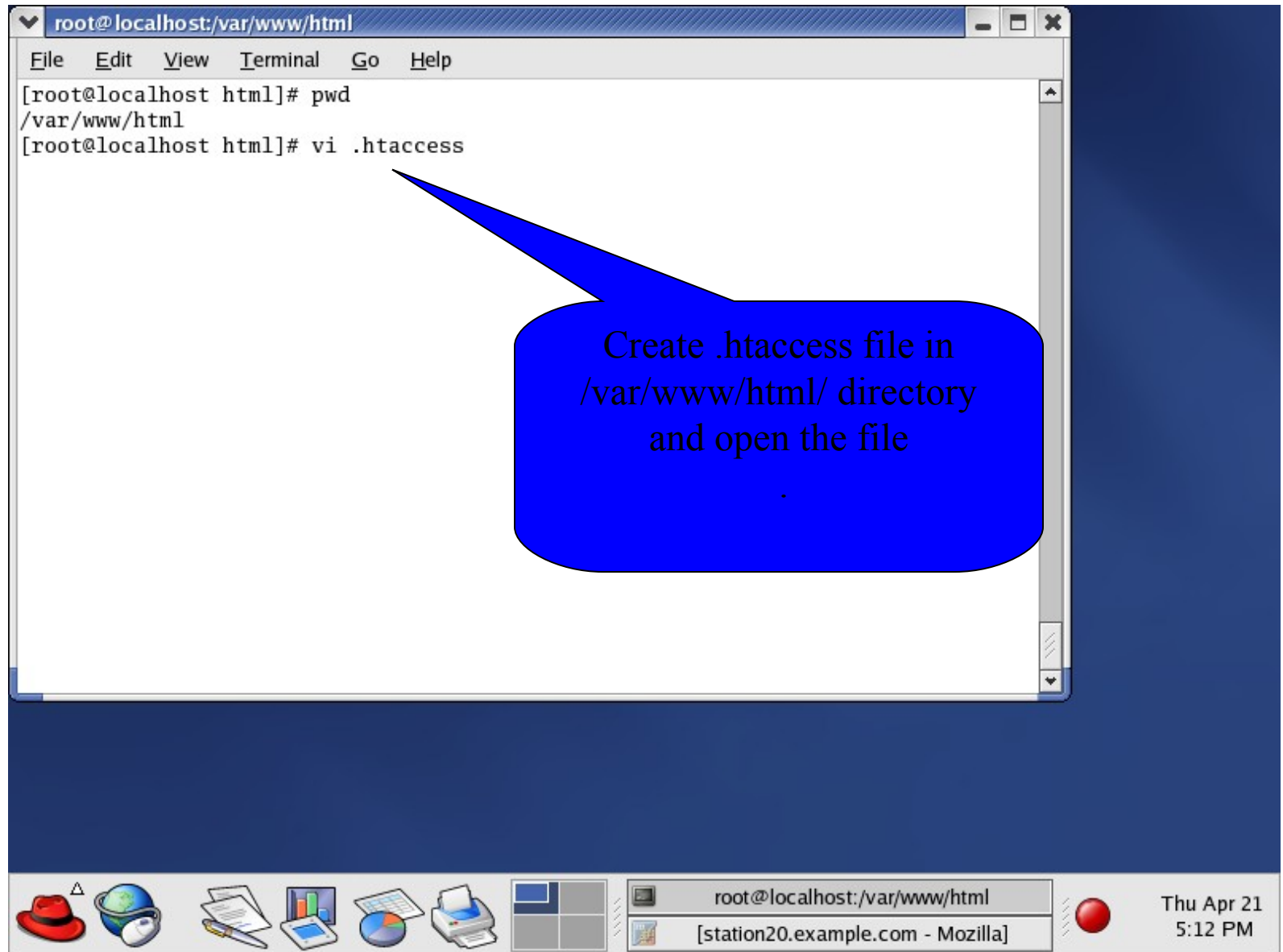
  <Directory /var/www/html>
    AllowOverride AuthConfig
  </Directory>
</VirtualHost>
-- INSERT --
```

Directives that can be overridden by the entries in an .htaccess file.

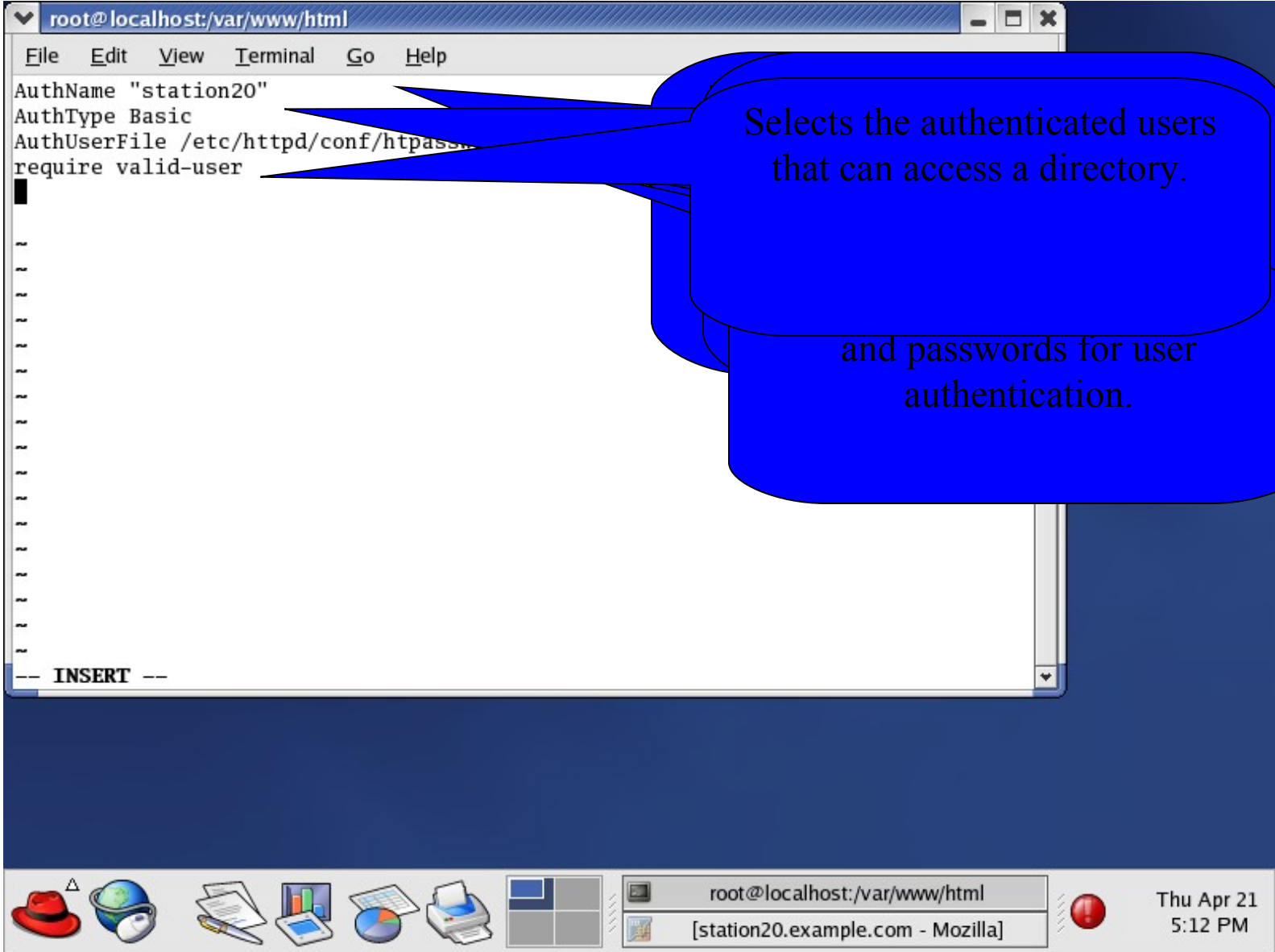
root@localhost:/var/www/html
[station20.example.com - Mozilla]

Thu Apr 21 5:06 PM

Web server configuration



Web server configuration



The image shows a terminal window titled `root@localhost:/var/www/html` with a menu bar (File, Edit, View, Terminal, Go, Help). The terminal displays the following configuration lines:

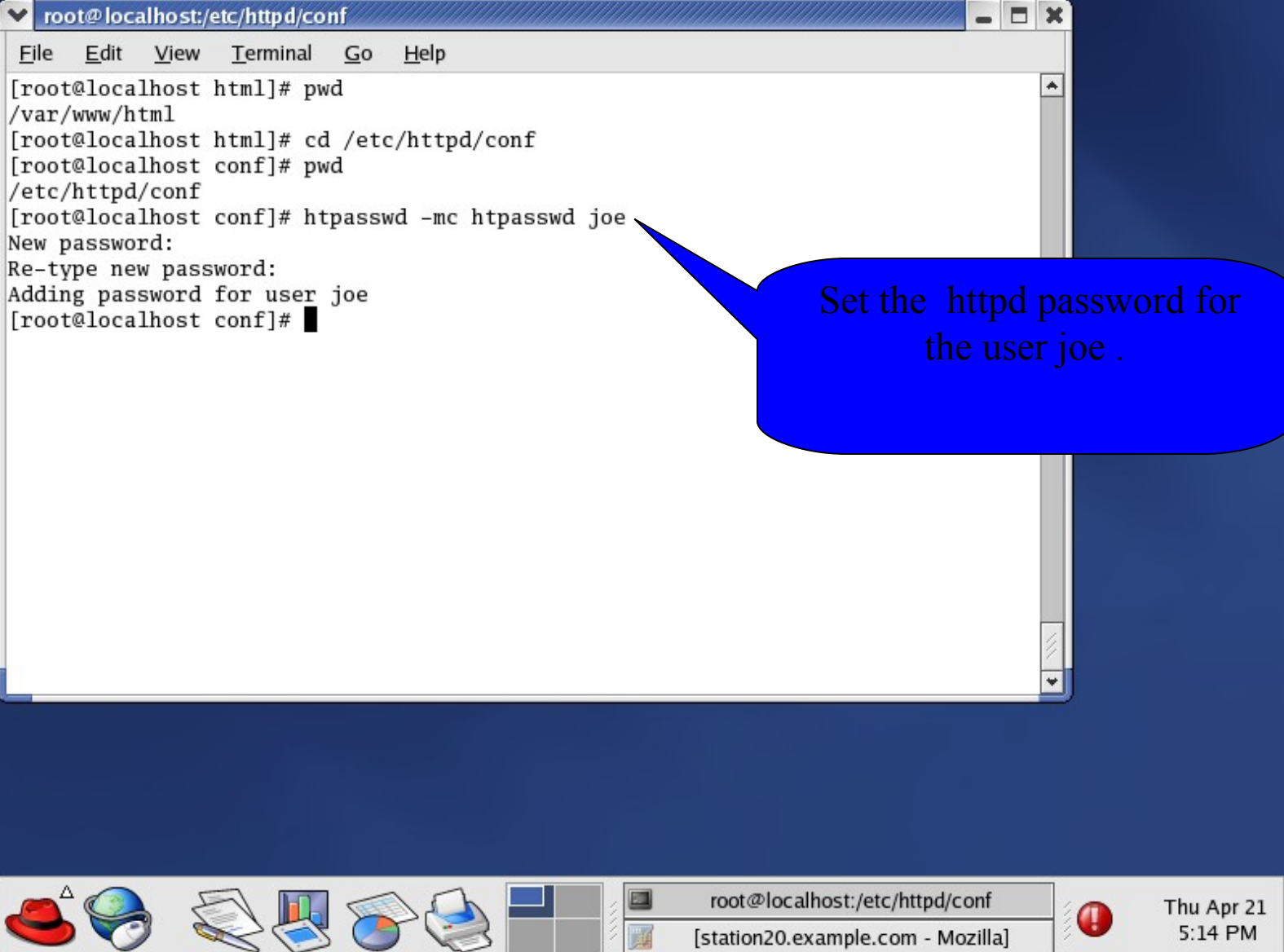
```
AuthName "station20"  
AuthType Basic  
AuthUserFile /etc/httpd/conf/htpasswd  
require valid-user
```

Two blue callout boxes provide explanations for the configuration:

- The first callout points to `AuthName "station20"` and contains the text: "Selects the authenticated users that can access a directory."
- The second callout points to `AuthUserFile /etc/httpd/conf/htpasswd` and contains the text: "and passwords for user authentication."

The terminal window also shows a series of tilde characters (~) and an `INSERT` status at the bottom left. The desktop environment at the bottom includes a taskbar with various icons (a red hat, a globe, a notepad, a bar chart, a pie chart, a printer, and a window icon) and a system tray showing the current session (`root@localhost:/var/www/html`), an open application (`[station20.example.com - Mozilla]`), and the date/time (`Thu Apr 21 5:12 PM`).

Web server configuration



The image shows a terminal window titled 'root@localhost:/etc/httpd/conf'. The terminal output is as follows:

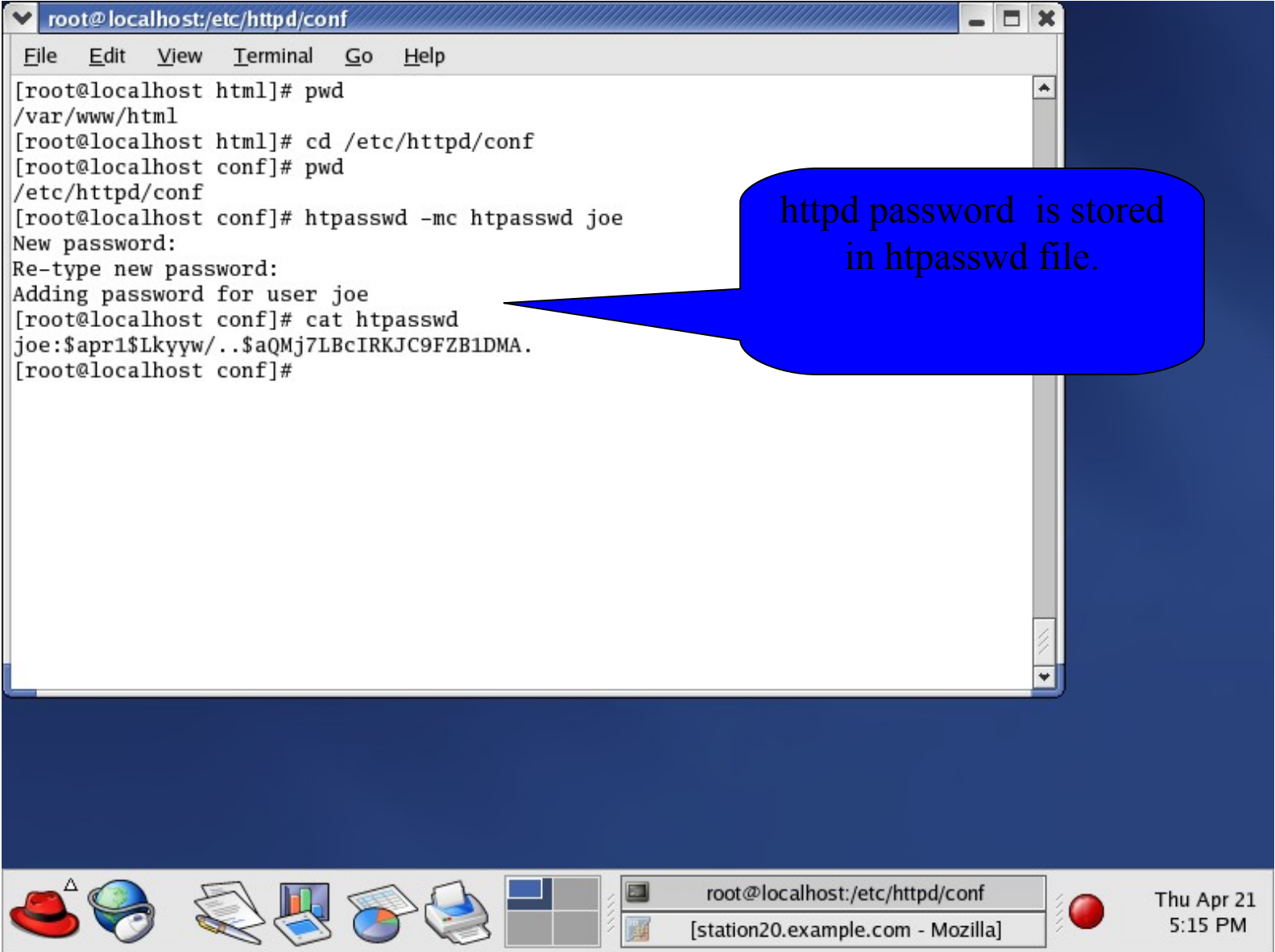
```
[root@localhost html]# pwd
/var/www/html
[root@localhost html]# cd /etc/httpd/conf
[root@localhost conf]# pwd
/etc/httpd/conf
[root@localhost conf]# htpasswd -mc htpasswd joe
New password:
Re-type new password:
Adding password for user joe
[root@localhost conf]#
```

A blue callout bubble points to the `htpasswd -mc htpasswd joe` command with the text: "Set the httpd password for the user joe .".

The desktop environment at the bottom includes a taskbar with various icons (a red hat, a globe, a notepad, a bar chart, a pie chart, a printer, and a window icon), a status bar showing the current directory and browser information, and a system clock.

root@localhost:/etc/httpd/conf
[station20.example.com - Mozilla]
Thu Apr 21 5:14 PM

Web server configuration



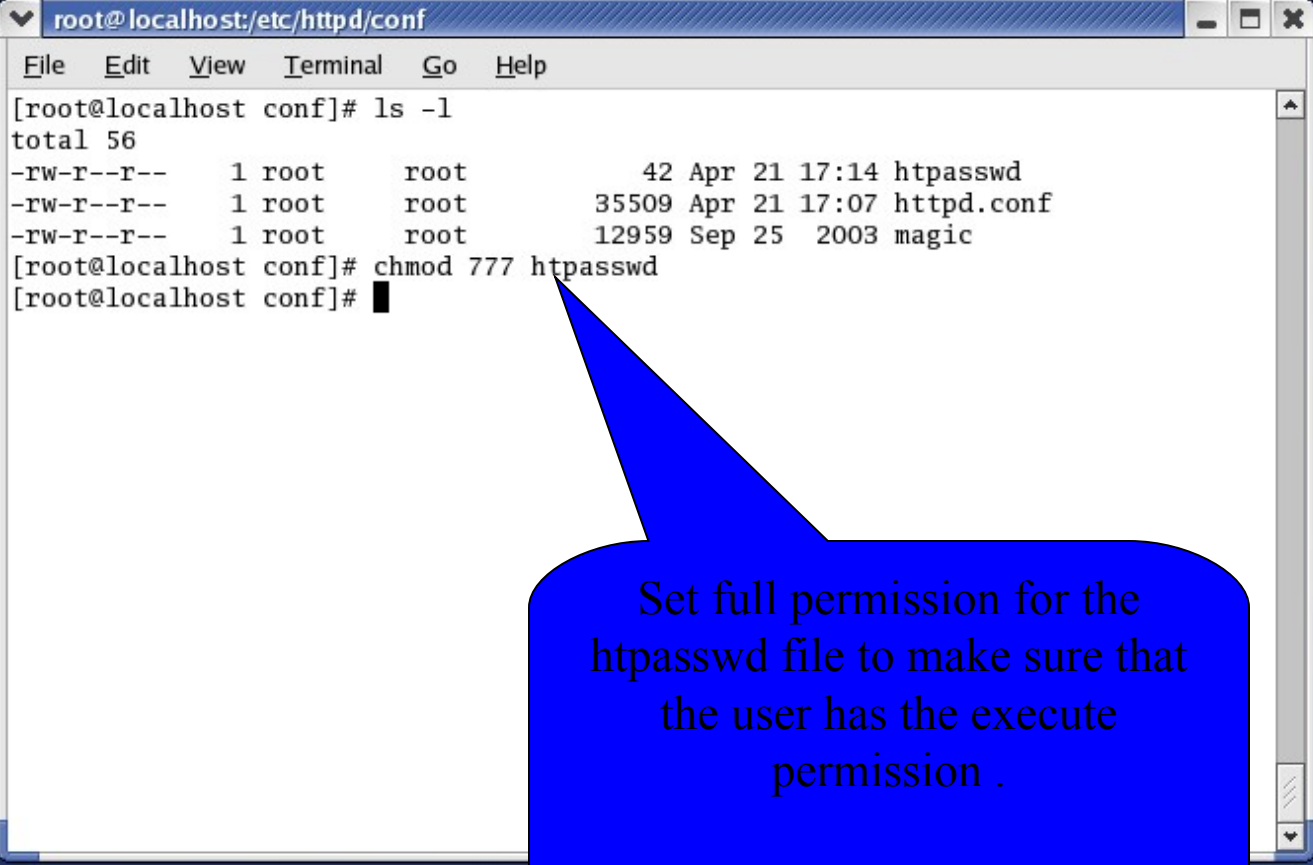
```
root@localhost:/etc/httpd/conf
File Edit View Terminal Go Help
[root@localhost html]# pwd
/var/www/html
[root@localhost html]# cd /etc/httpd/conf
[root@localhost conf]# pwd
/etc/httpd/conf
[root@localhost conf]# htpasswd -mc htpasswd joe
New password:
Re-type new password:
Adding password for user joe
[root@localhost conf]# cat htpasswd
joe:$apr1$Lkyyw/..$aQMj7LBcIRKJC9FZB1DMA.
[root@localhost conf]#
```

httpd password is stored in htpasswd file.

root@localhost:/etc/httpd/conf
[station20.example.com - Mozilla]

Thu Apr 21 5:15 PM

Web server configuration



The terminal window shows the following output for the command `ls -l`:

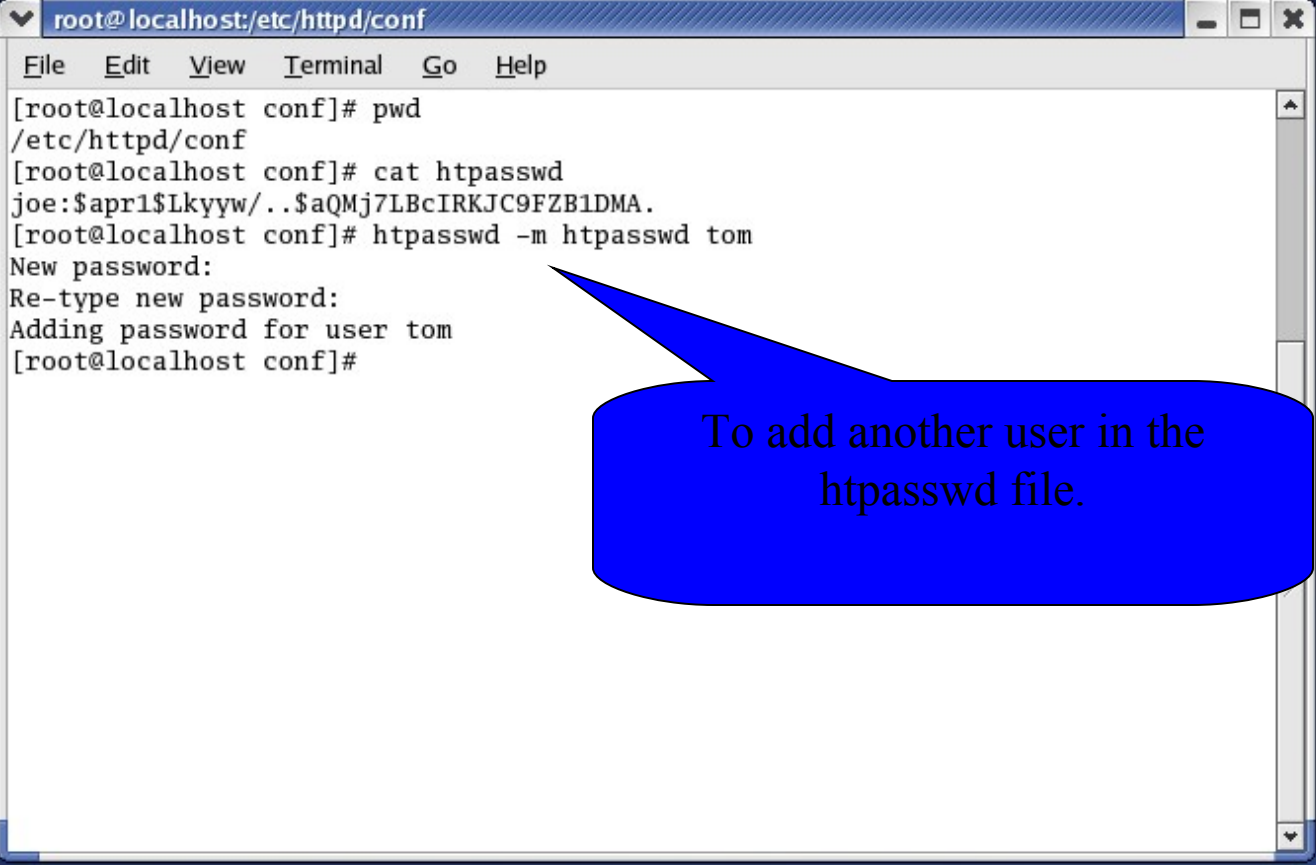
```
[root@localhost conf]# ls -l
total 56
-rw-r--r--  1 root    root      42 Apr 21 17:14 htpasswd
-rw-r--r--  1 root    root    35509 Apr 21 17:07 httpd.conf
-rw-r--r--  1 root    root    12959 Sep 25  2003 magic
```

The user then executes `chmod 777 htpasswd`. A blue callout bubble points to this command with the text:

Set full permission for the htpasswd file to make sure that the user has the execute permission .

The desktop taskbar at the bottom includes icons for a red hat, a globe, documents, a bar chart, a pie chart, a printer, and a window manager. The system tray shows the terminal path `root@localhost:/etc/httpd/conf`, the browser `[station20.example.com - Mozilla]`, and the date/time `Thu Apr 21 5:15 PM`.

Web server configuration



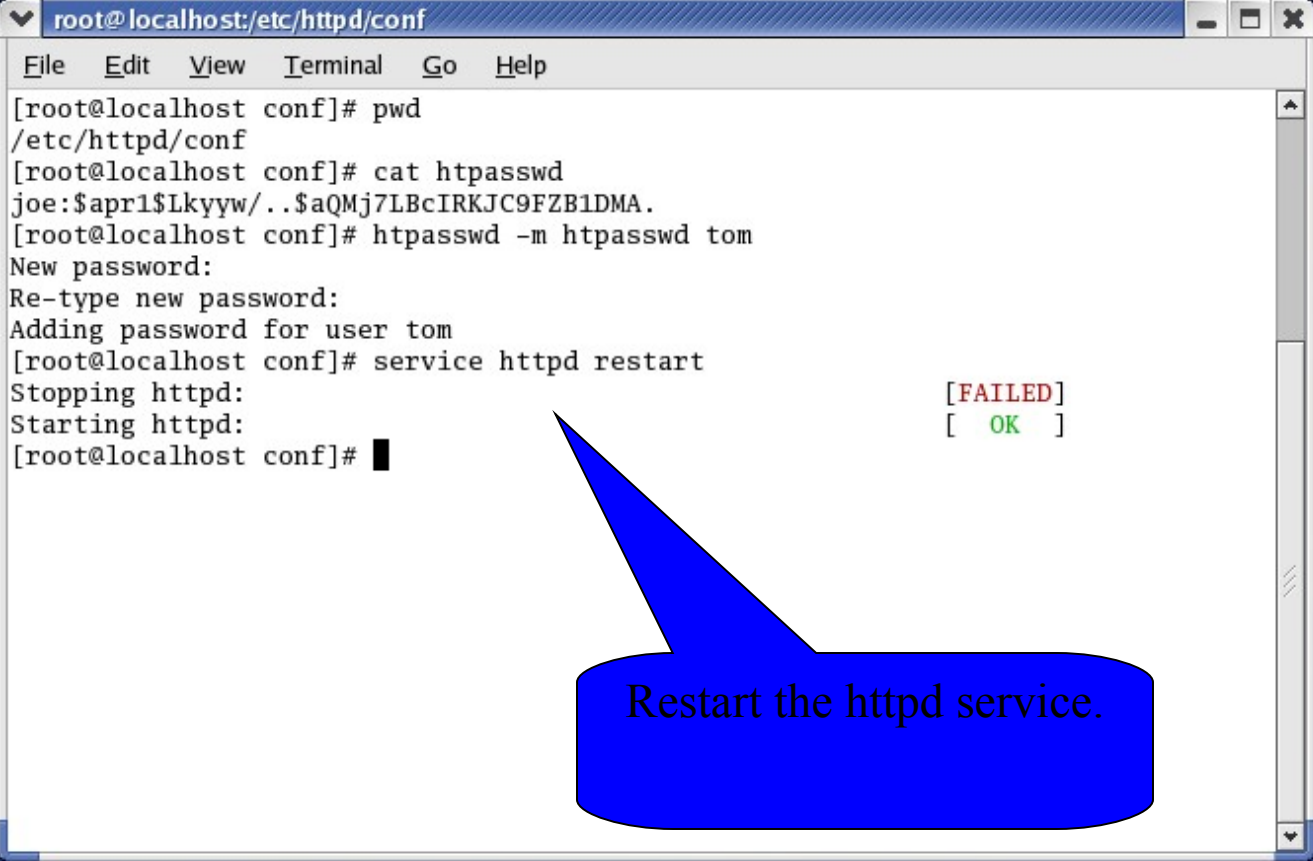
The image shows a terminal window titled 'root@localhost:/etc/httpd/conf'. The terminal output is as follows:

```
[root@localhost conf]# pwd
/etc/httpd/conf
[root@localhost conf]# cat htpasswd
joe:$apr1$Lkyyw/..$aQMj7LBcIRKJC9FZB1DMA.
[root@localhost conf]# htpasswd -m htpasswd tom
New password:
Re-type new password:
Adding password for user tom
[root@localhost conf]#
```

A blue callout bubble points to the `htpasswd -m htpasswd tom` command with the text: "To add another user in the htpasswd file."

The desktop environment at the bottom includes a taskbar with icons for a red hat, a globe, a notepad, a bar chart, a pie chart, a printer, and a window manager. The system tray shows the current path 'root@localhost:/etc/httpd/conf' and the date/time 'Thu Apr 21 5:21 PM'.

Web server configuration



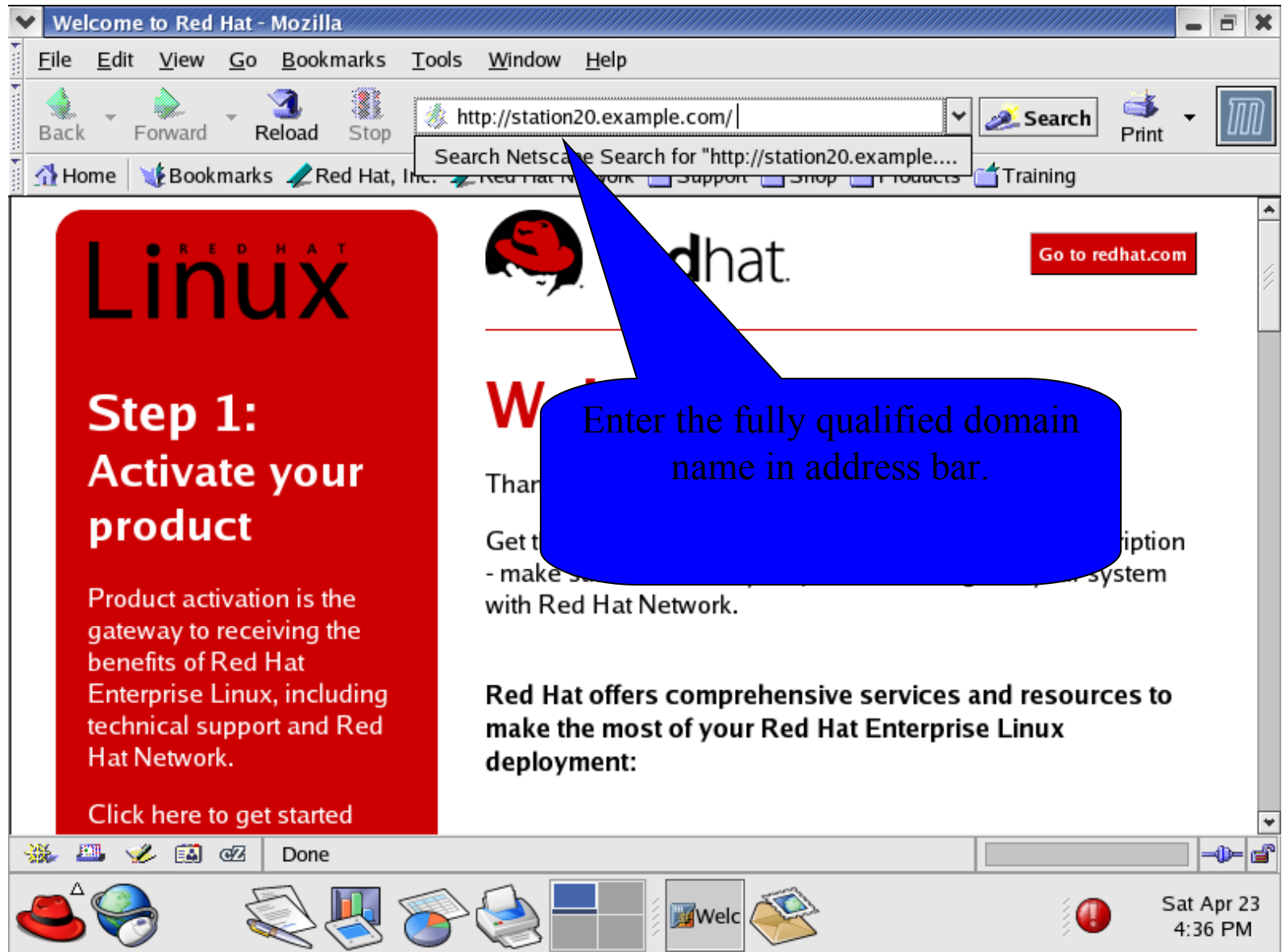
A terminal window titled 'root@localhost:/etc/httpd/conf' with a menu bar (File, Edit, View, Terminal, Go, Help). The terminal shows the following commands and output:

```
[root@localhost conf]# pwd
/etc/httpd/conf
[root@localhost conf]# cat htpasswd
joe:$apr1$Lkyyw/..$aQMj7LBcIRKJC9FZB1DMA.
[root@localhost conf]# htpasswd -m htpasswd tom
New password:
Re-type new password:
Adding password for user tom
[root@localhost conf]# service httpd restart
Stopping httpd:
Starting httpd:
[FAILED]
[ OK ]
```

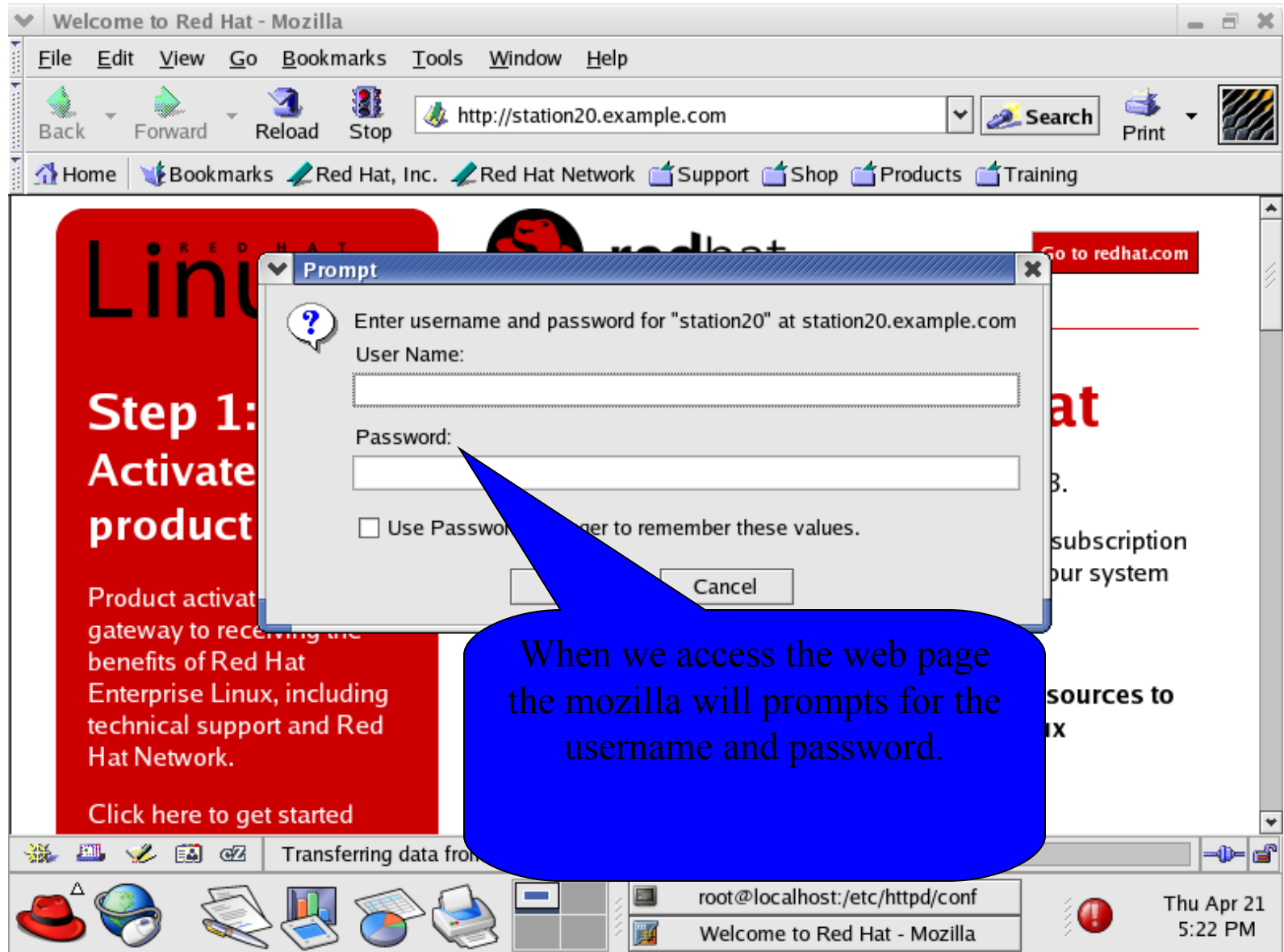
A blue callout bubble points to the 'Starting httpd:' line with the text: "Restart the httpd service."

The desktop environment at the bottom includes a taskbar with icons for a red hat, a globe, a notepad, a bar chart, a pie chart, a printer, and a window manager. The system tray shows the terminal path 'root@localhost:/etc/httpd/conf' and the date/time 'Thu Apr 21 5:22 PM'.

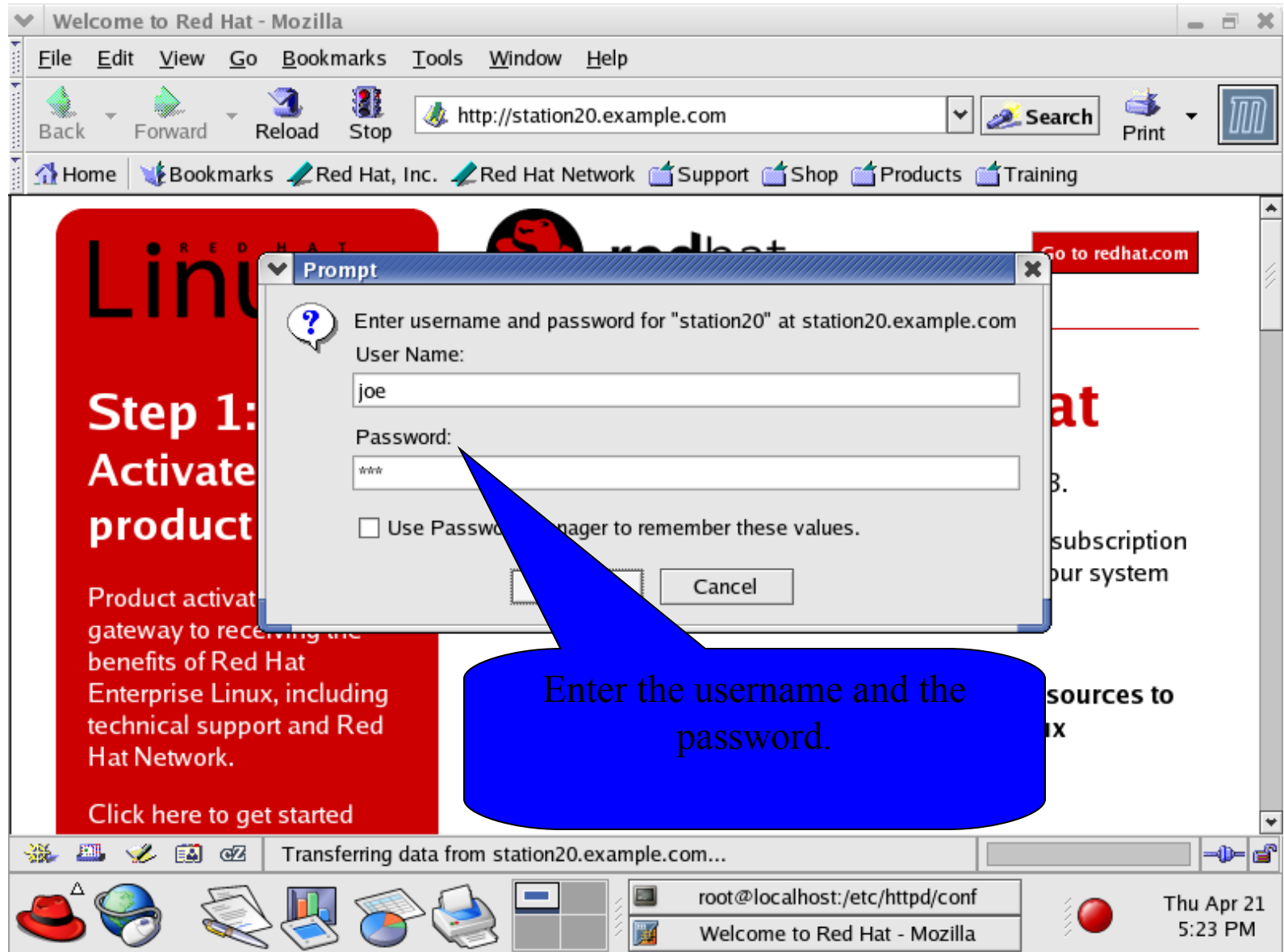
Web server configuration



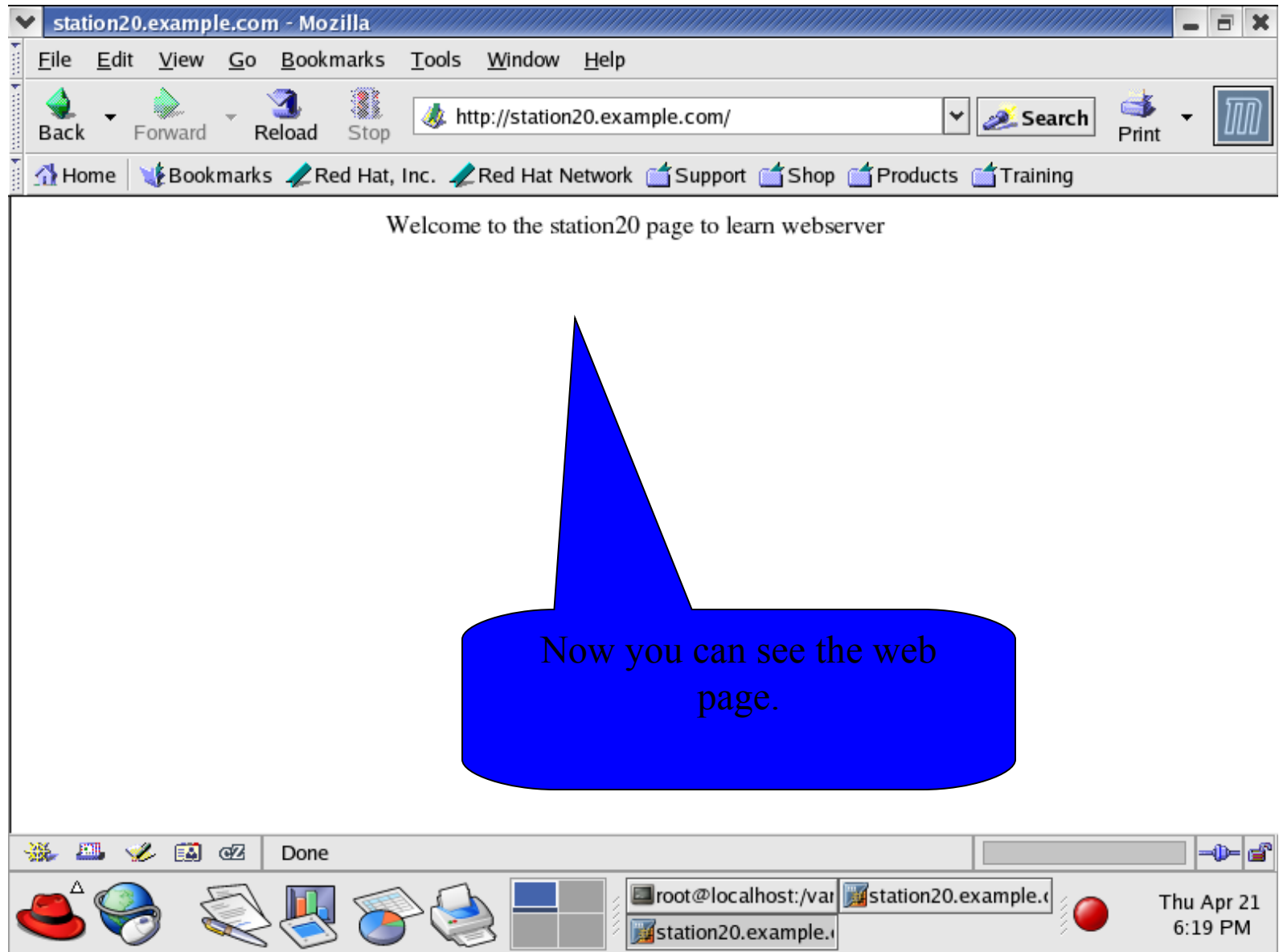
Web server configuration



Web server configuration



Web server configuration

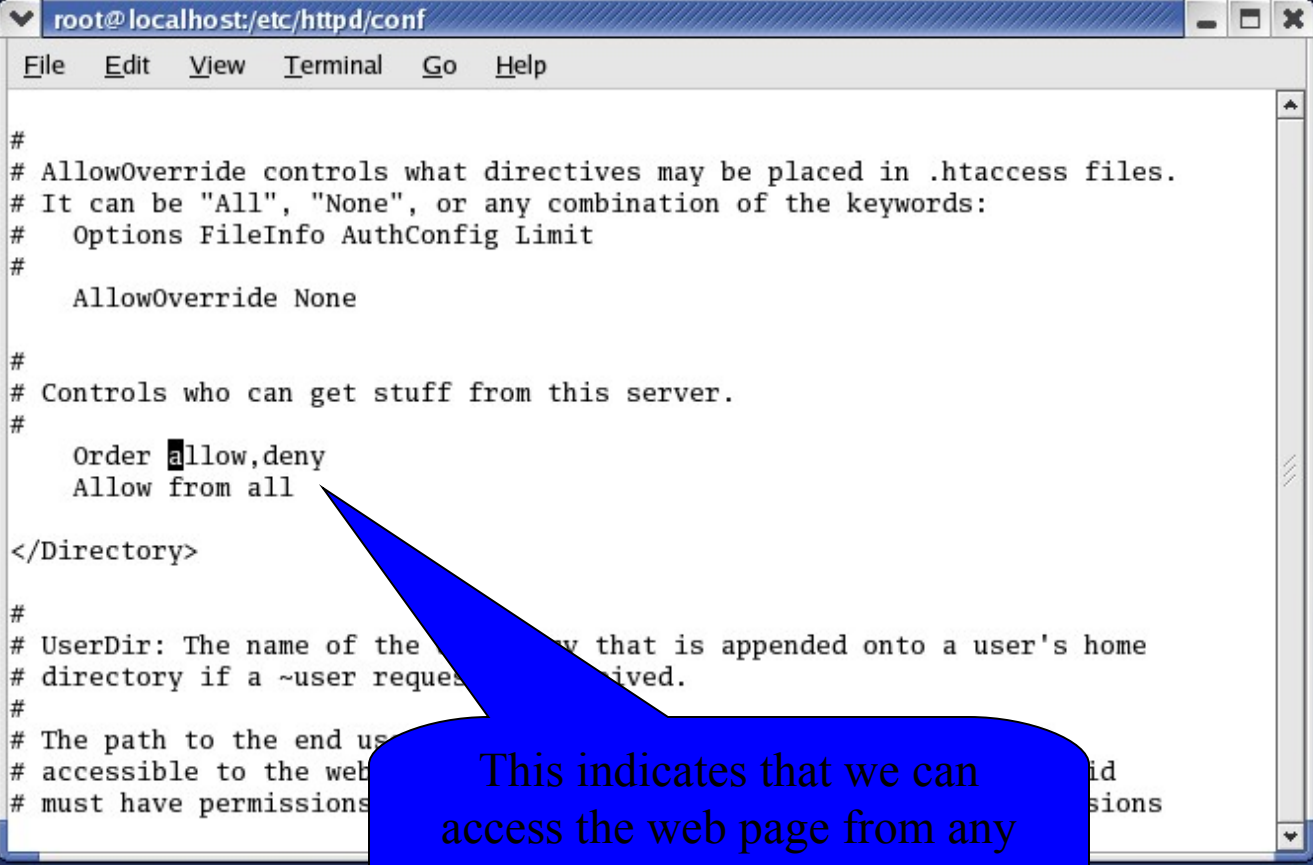


Access Control

- The web access can be controlled by using access control directives such as allow and deny.

Directive	Description
deny from host host	Determines hosts that can access a given directory: all, or partial or full domain name or ip address.
allow from host host	Determines which hosts can access a given directory: all, or partial or full domain name or ip address.

Web server configuration



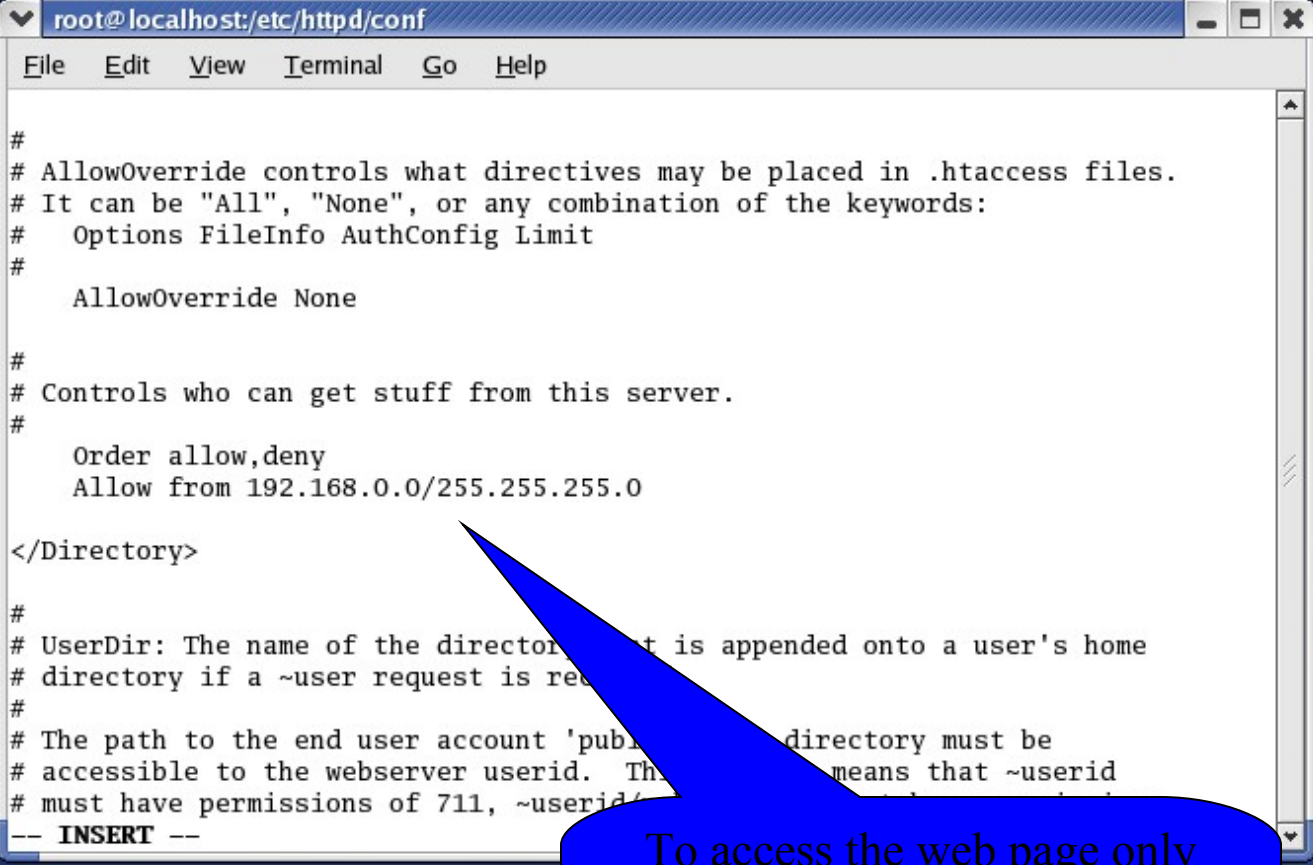
The screenshot shows a terminal window titled 'root@localhost:/etc/httpd/conf'. The terminal displays the contents of an Apache configuration file, likely httpd.conf. The visible text includes:

```
#  
# AllowOverride controls what directives may be placed in .htaccess files.  
# It can be "All", "None", or any combination of the keywords:  
#   Options FileInfo AuthConfig Limit  
#  
#   AllowOverride None  
  
#  
# Controls who can get stuff from this server.  
#  
#   Order allow,deny  
#   Allow from all  
  
</Directory>  
  
#  
# UserDir: The name of the directory that is appended onto a user's home  
# directory if a ~user request is received.  
#  
# The path to the end user's home directory must be accessible to the web  
# must have permissions
```

A blue callout bubble points to the 'Order allow,deny' line, containing the text: 'This indicates that we can access the web page from any network.'

The terminal window has a menu bar with 'File', 'Edit', 'View', 'Terminal', 'Go', and 'Help'. The taskbar at the bottom shows various icons (a red hat, a globe, a notepad, a bar chart, a pie chart, a printer, and a window icon) and a status bar with the text 'root@localhost:/etc/httpd/conf', '[station20.example.com - Mozilla]', and the date/time 'Thu Apr 21 5:41 PM'.

Web server configuration



```
root@localhost:/etc/httpd/conf
File Edit View Terminal Go Help

#
# AllowOverride controls what directives may be placed in .htaccess files.
# It can be "All", "None", or any combination of the keywords:
#   Options FileInfo AuthConfig Limit
#
#   AllowOverride None
#
# Controls who can get stuff from this server.
#
#   Order allow,deny
#   Allow from 192.168.0.0/255.255.255.0
#
</Directory>
#
# UserDir: The name of the directory is appended onto a user's home
# directory if a ~user request is received.
#
# The path to the end user account 'public_html' directory must be
# accessible to the webserver userid. This means that ~userid
# must have permissions of 711, ~userid/public_html must have
# permissions of 711, and the directory must be owned by ~userid.
-- INSERT --
```

To access the web page only from the above specified network.

root@localhost:/etc/httpd/conf
[station20.example.com - Mozilla]

Thu Apr 21 5:42 PM

Virtual host

- Virtual hosts allow you to host multiple domains from a single machine.
- Apache supports two different type of virtual hosts.
-
- Name-based virtual hosts
- Ip-based virtual hosts
-

Ip and Named-based Virtual Hosts

- IP-based virtual hosts
 - -Ip-based virtual hosts are used when a single machine has several different IP address.
 - -To run a virtual host on each IP address.
- Named-based virtual hosts
 - -It allow you to host many virtual hosts from a single IP address.

Name-based virtual hosts

The terminal window shows the following configuration:

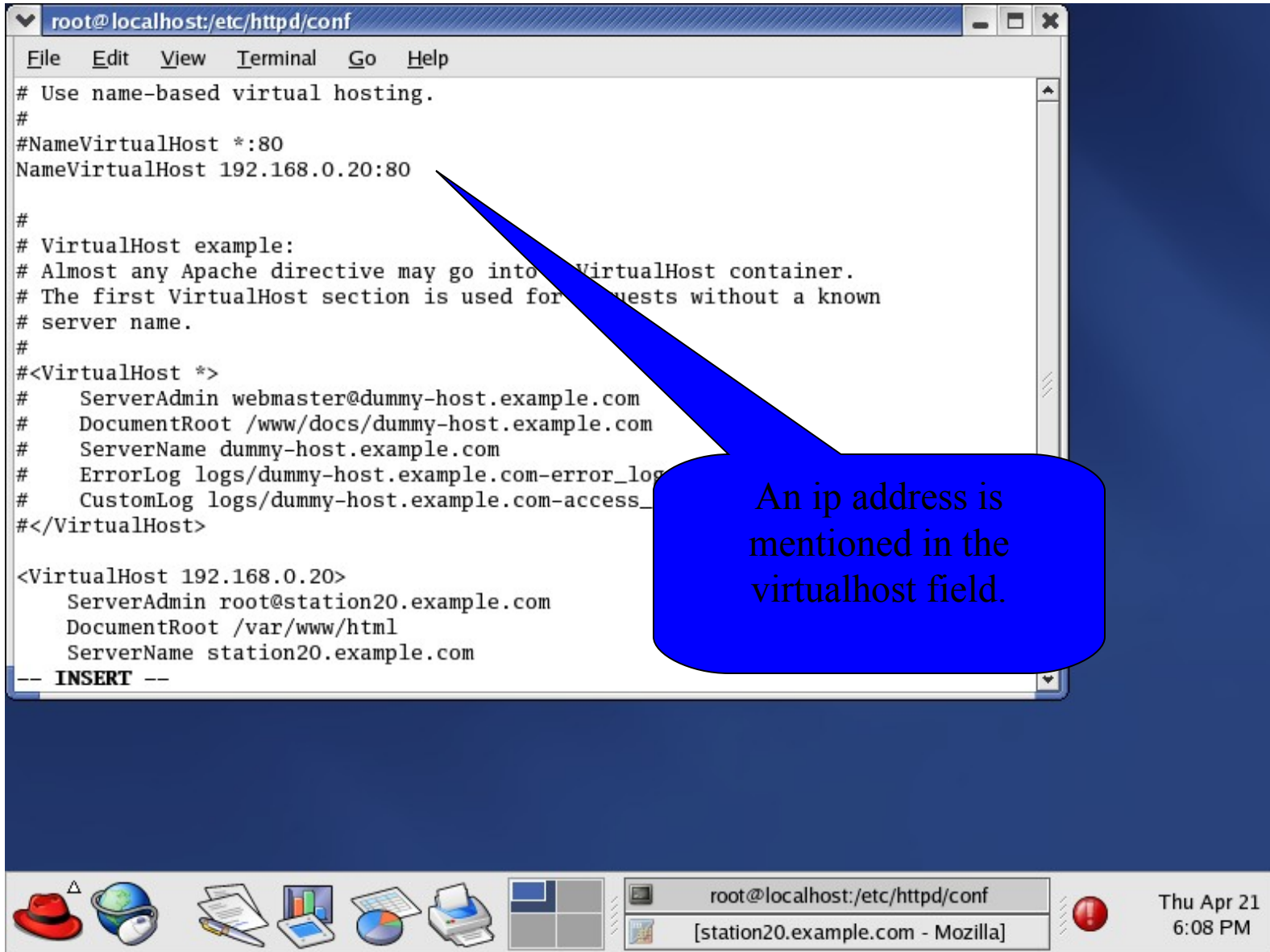
```
#<VirtualHost *>
#   ServerAdmin webmaster@dummy-host.example.com
#   DocumentRoot /www/docs/dummy-host.example.com
#   ServerName dummy-host.example.com
#   ErrorLog logs/dummy-host.example.com-error_log
#   CustomLog logs/dummy-host.example.com-access_log common
#</VirtualHost>

<VirtualHost 192.168.0.20>
  ServerAdmin root@station20.example.com
  DocumentRoot /var/www/html
  ServerName station20.example.com
  ErrorLog logs/station20.example.com-error_log
  CustomLog logs/station20.example.com-access_log common
</VirtualHost>
<VirtualHost 192.168.0.20>
  ServerAdmin root@station20.test.com
  DocumentRoot /var/www/html1
  ServerName station20.test.com
  ErrorLog logs/station20.test.com-error_log
  CustomLog logs/station20.test.com-access_log common
</VirtualHost>
```

Two domains are used in the name of example and test.com in a single ip address 192.168.0.20.

The terminal window has a title bar "root@localhost:~" and a menu bar "File Edit View Terminal Go Help". The taskbar at the bottom shows various icons including a red hat, a globe, a printer, and a folder, along with the date "Wed Apr 27 11:53 AM".

Name-based virtual hosts

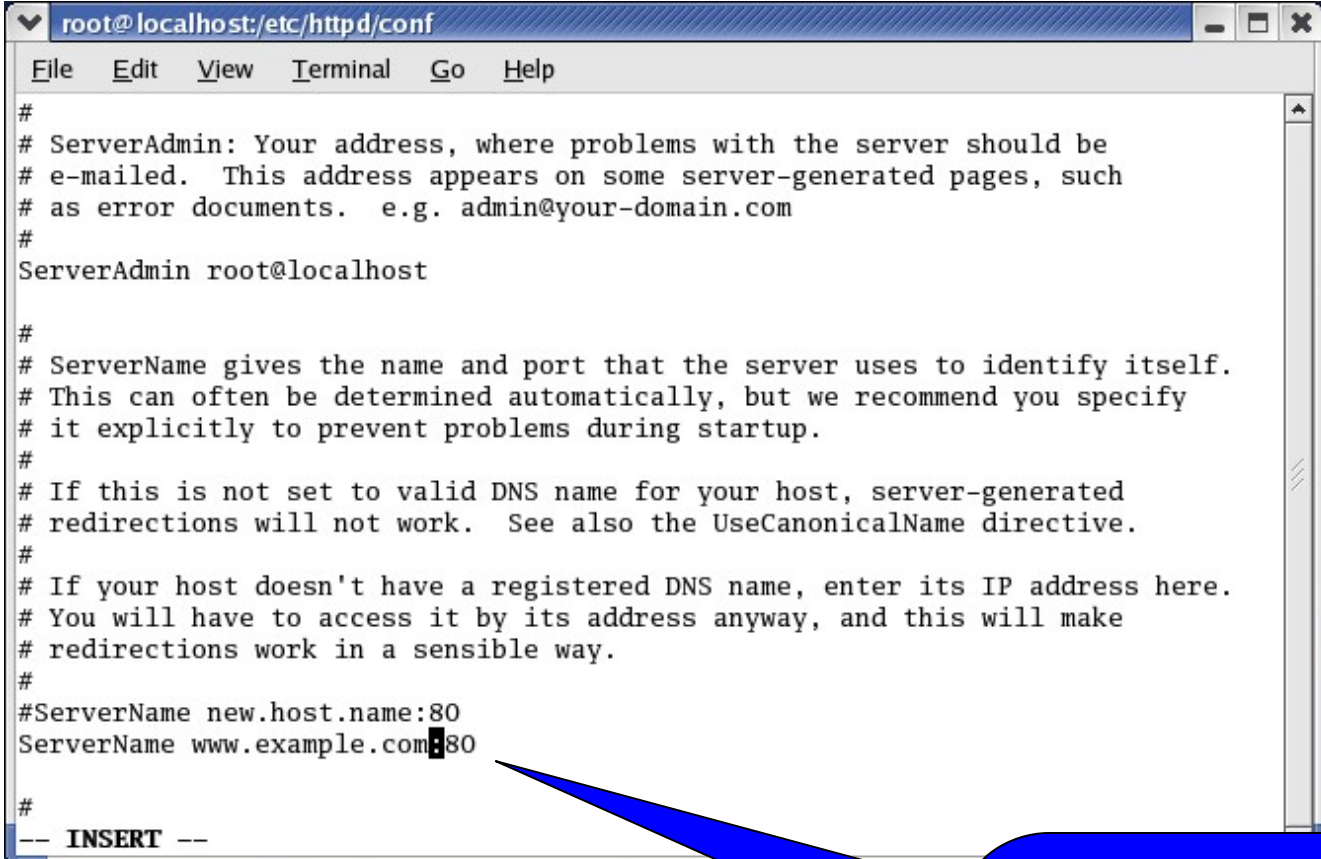


The screenshot shows a text editor window titled 'root@localhost:/etc/httpd/conf' displaying the configuration file 'httpd.conf'. The file contains comments and directives for name-based virtual hosting. A blue callout bubble points to the 'NameVirtualHost' directive, stating: 'An ip address is mentioned in the virtualhost field.'

```
root@localhost:/etc/httpd/conf
File Edit View Terminal Go Help
# Use name-based virtual hosting.
#
#NameVirtualHost *:80
NameVirtualHost 192.168.0.20:80
#
# VirtualHost example:
# Almost any Apache directive may go into a VirtualHost container.
# The first VirtualHost section is used for requests without a known
# server name.
#
#<VirtualHost *>
#   ServerAdmin webmaster@dummy-host.example.com
#   DocumentRoot /www/docs/dummy-host.example.com
#   ServerName dummy-host.example.com
#   ErrorLog logs/dummy-host.example.com-error_log
#   CustomLog logs/dummy-host.example.com-access_log
#</VirtualHost>
<VirtualHost 192.168.0.20>
    ServerAdmin root@station20.example.com
    DocumentRoot /var/www/html
    ServerName station20.example.com
-- INSERT --
```

root@localhost:/etc/httpd/conf
[station20.example.com - Mozilla]
Thu Apr 21 6:08 PM

Name-based virtual hosts



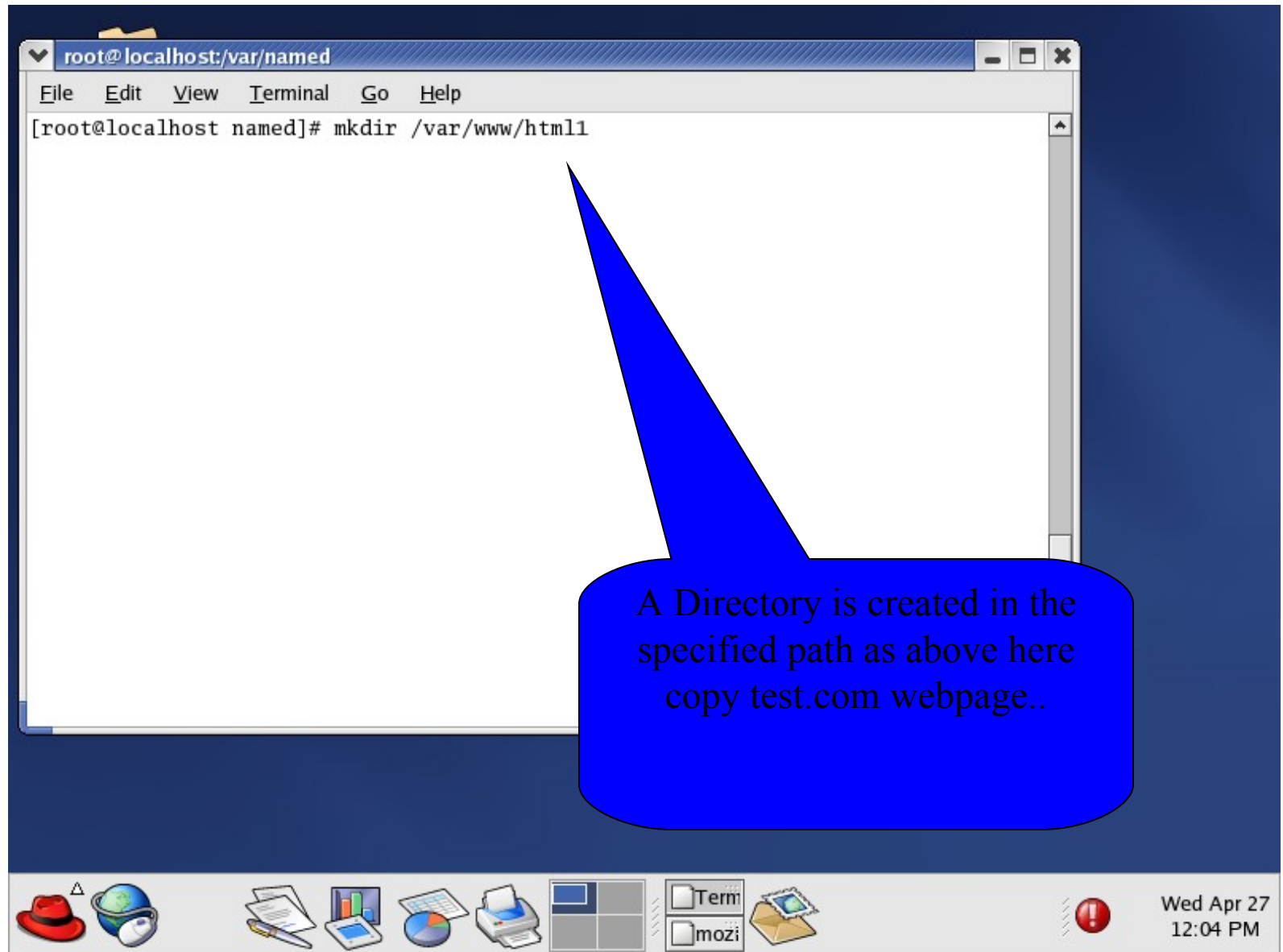
The screenshot shows a text editor window titled 'root@localhost:/etc/httpd/conf'. The menu bar includes 'File', 'Edit', 'View', 'Terminal', 'Go', and 'Help'. The text content is as follows:

```
#  
# ServerAdmin: Your address, where problems with the server should be  
# e-mailed. This address appears on some server-generated pages, such  
# as error documents. e.g. admin@your-domain.com  
#  
ServerAdmin root@localhost  
  
#  
# ServerName gives the name and port that the server uses to identify itself.  
# This can often be determined automatically, but we recommend you specify  
# it explicitly to prevent problems during startup.  
#  
# If this is not set to valid DNS name for your host, server-generated  
# redirections will not work. See also the UseCanonicalName directive.  
#  
# If your host doesn't have a registered DNS name, enter its IP address here.  
# You will have to access it by its address anyway, and this will make  
# redirections work in a sensible way.  
#  
#ServerName new.host.name:80  
ServerName www.example.com:80  
  
#  
-- INSERT --
```

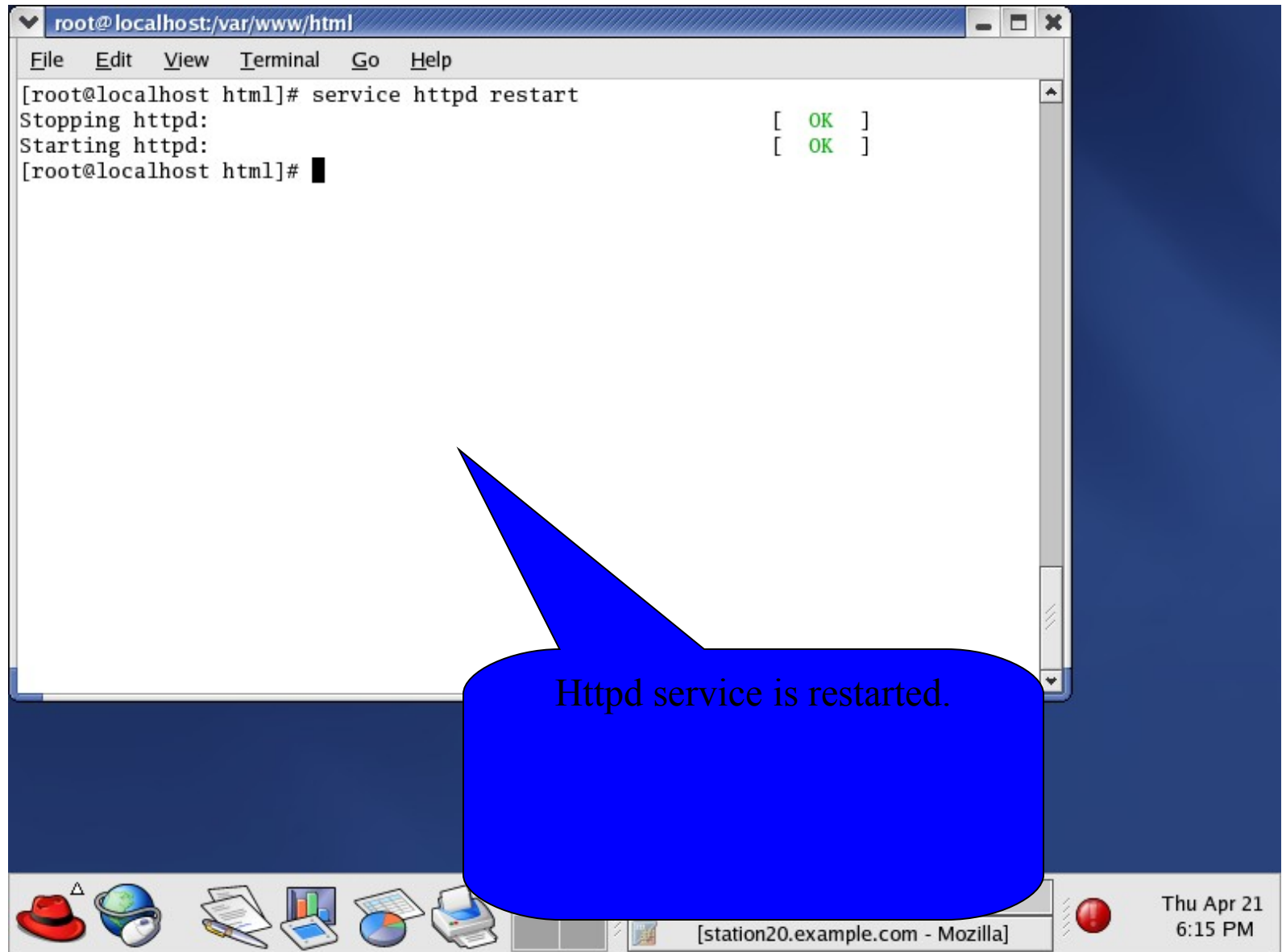
A blue callout bubble points to the line 'ServerName www.example.com:80' with the text: 'The server fully qualified domain name is described here.'

The taskbar at the bottom shows various icons including a red hat, a globe, a notepad, a bar chart, a pie chart, a printer, and a window icon. The system tray on the right shows the user 'root@lo', the file '[station20.ex', and the date/time 'Thu Apr 21 6:09 PM'.

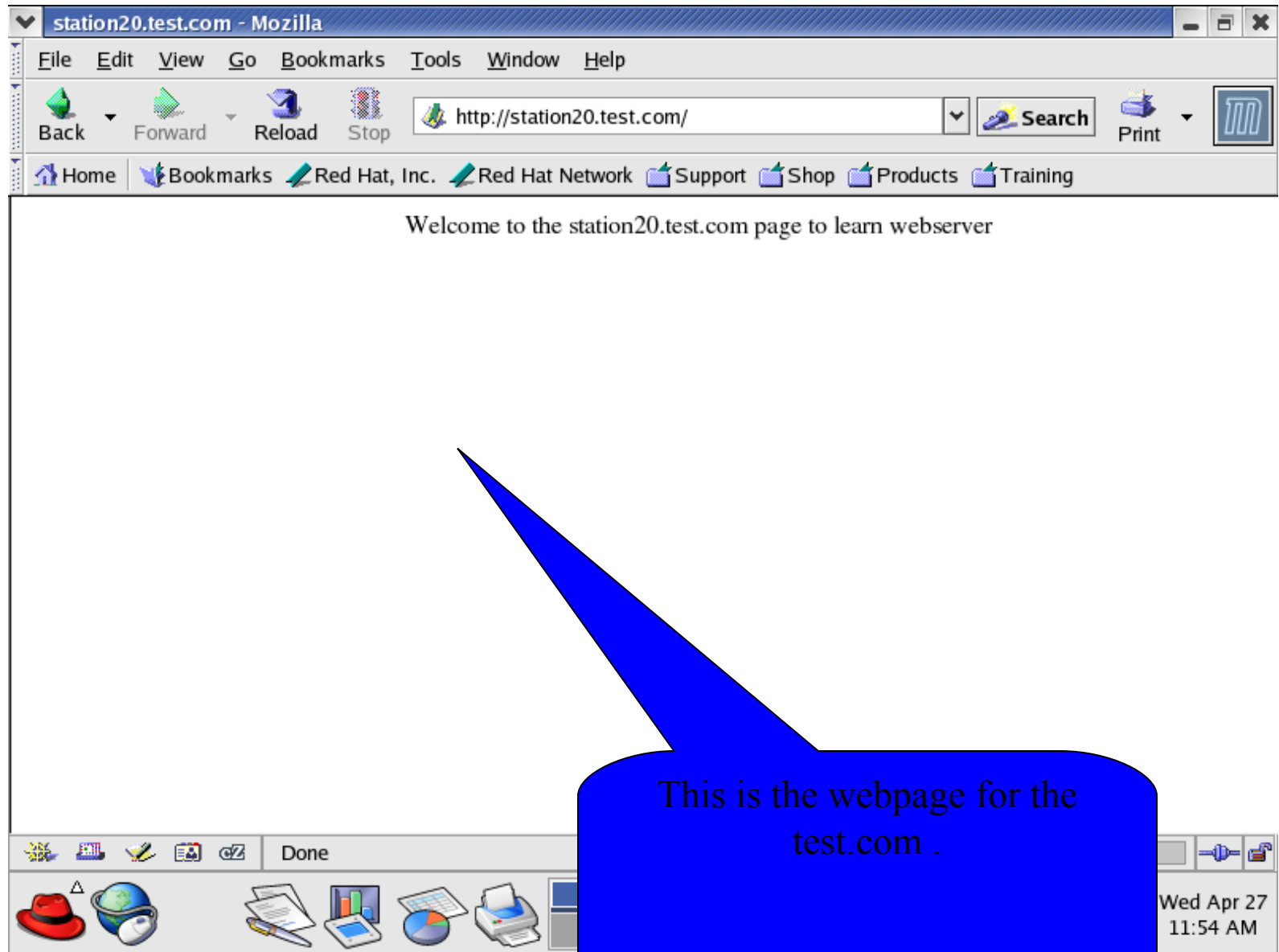
Name-based virtual hosts



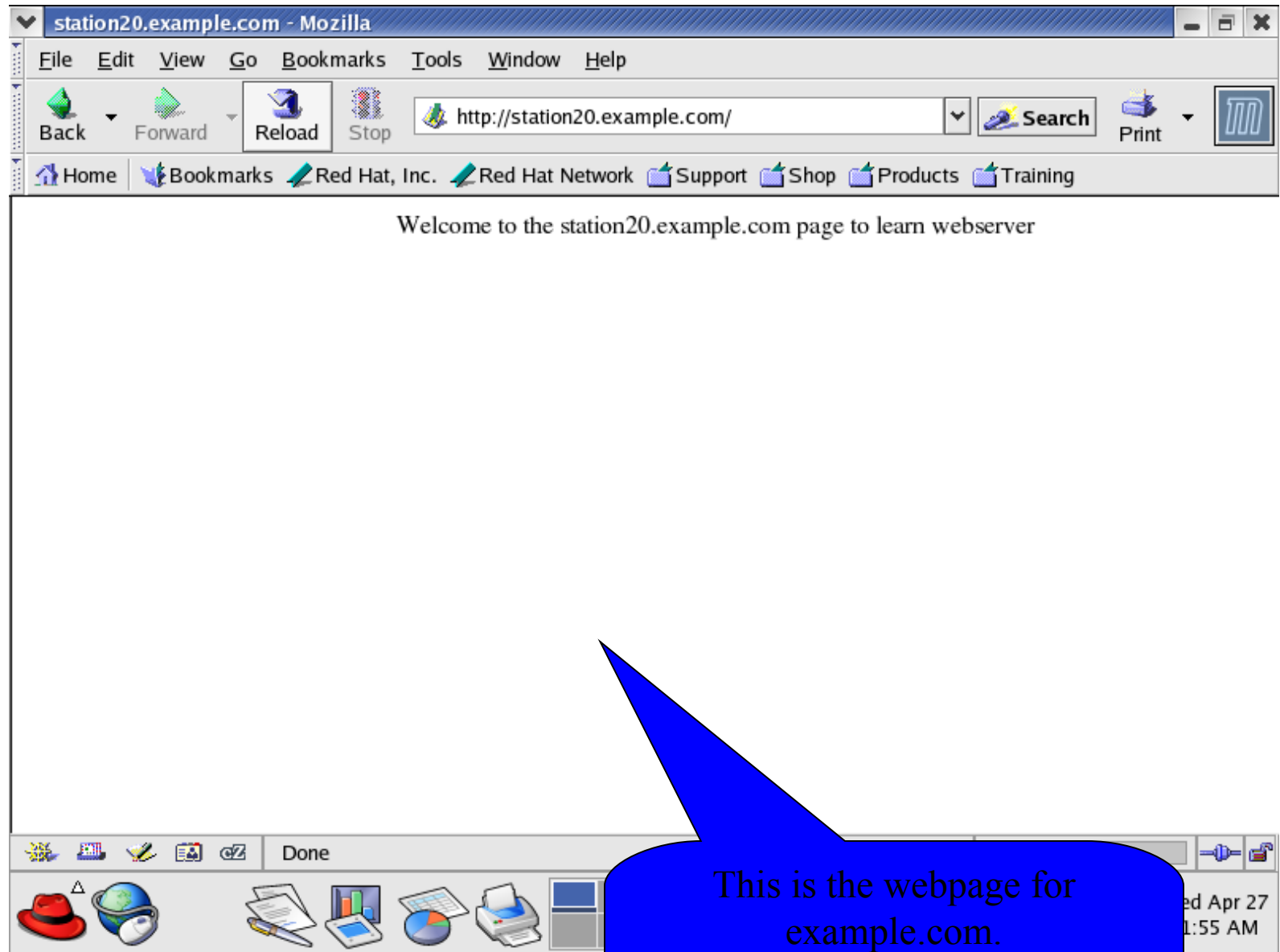
Name-based virtual hosts



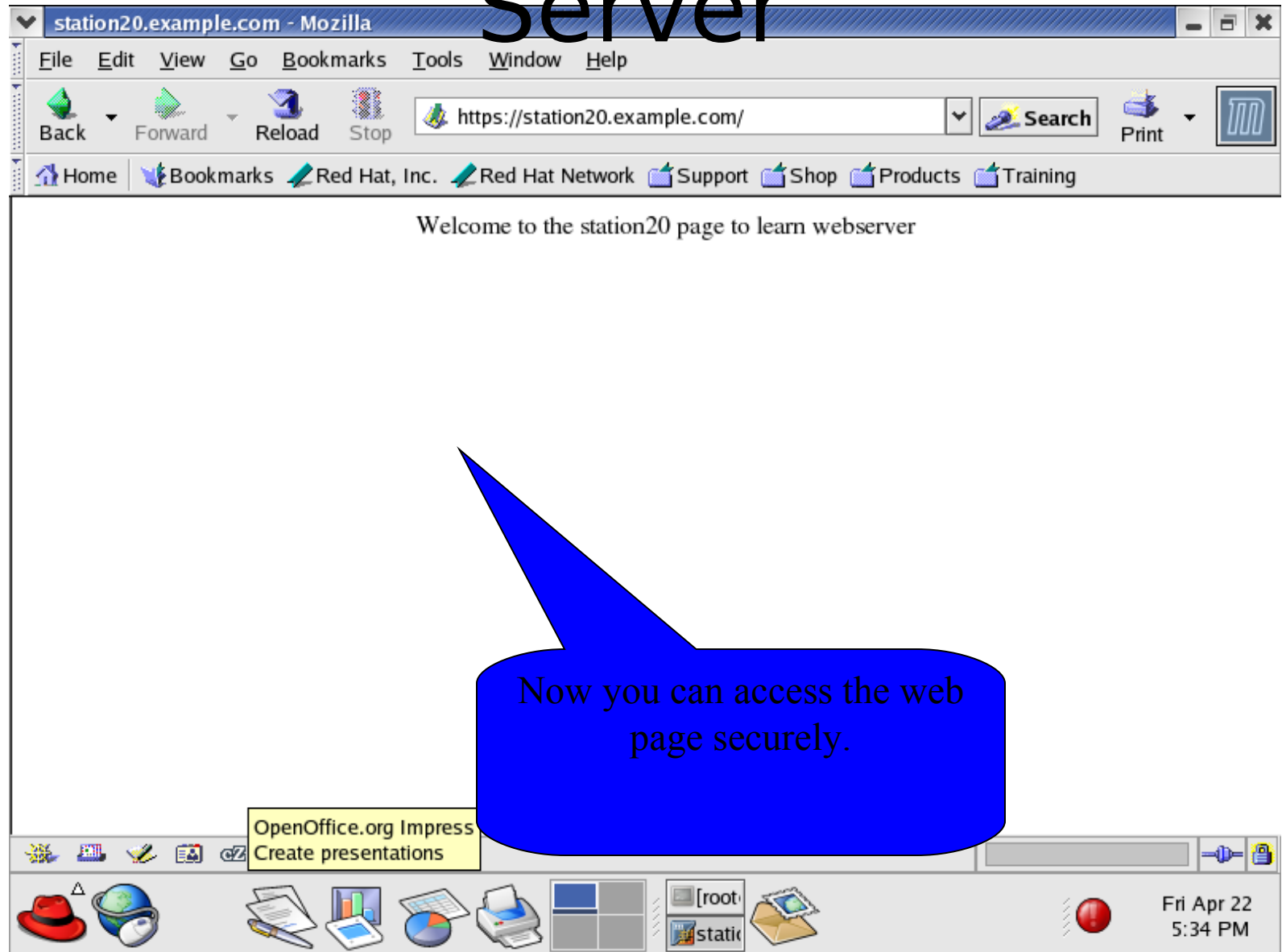
Name-based virtual hosts



Name-based virtual hosts



Apache Encrypted Web Server



Apache Encrypted Web Server

Apache can provide encrypted communications using the `mod_ssl` apache module.

-

To make use of encrypted communications, a client must request the `https` protocol.

-

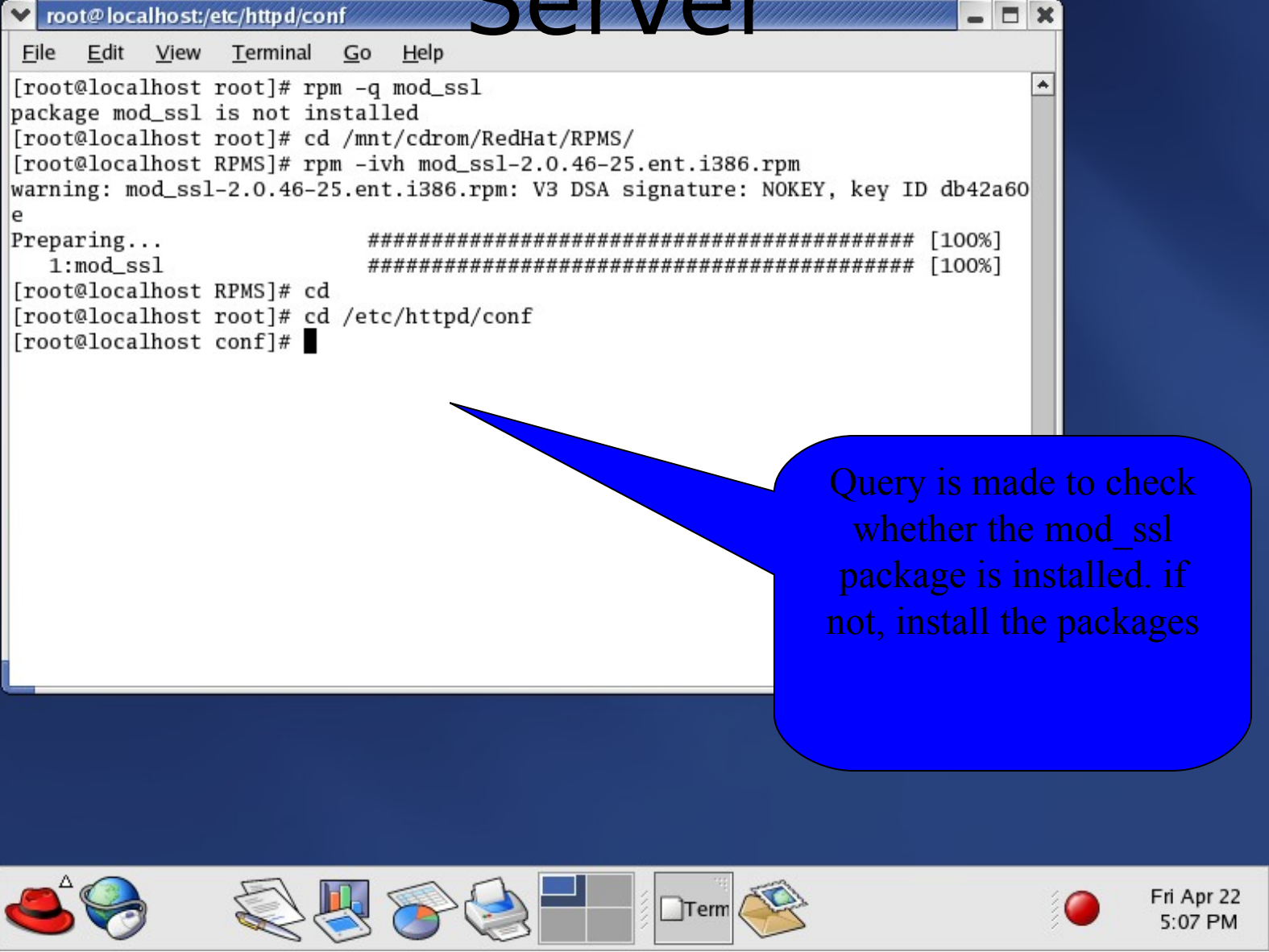
Encryption is based on either the RSA or DSA algorithm.

-

Apache Encrypted Web Server

- Package - `mod_ssl`
- Port number - `443`
- Certificate - `conf/ssl.crt/server.crt`
- Private key - `conf/ssl.key/server.key`
- Self-signed cert - `make testcert`
- Certificate signature req - `make certreq`

Apache Encrypted Web Server

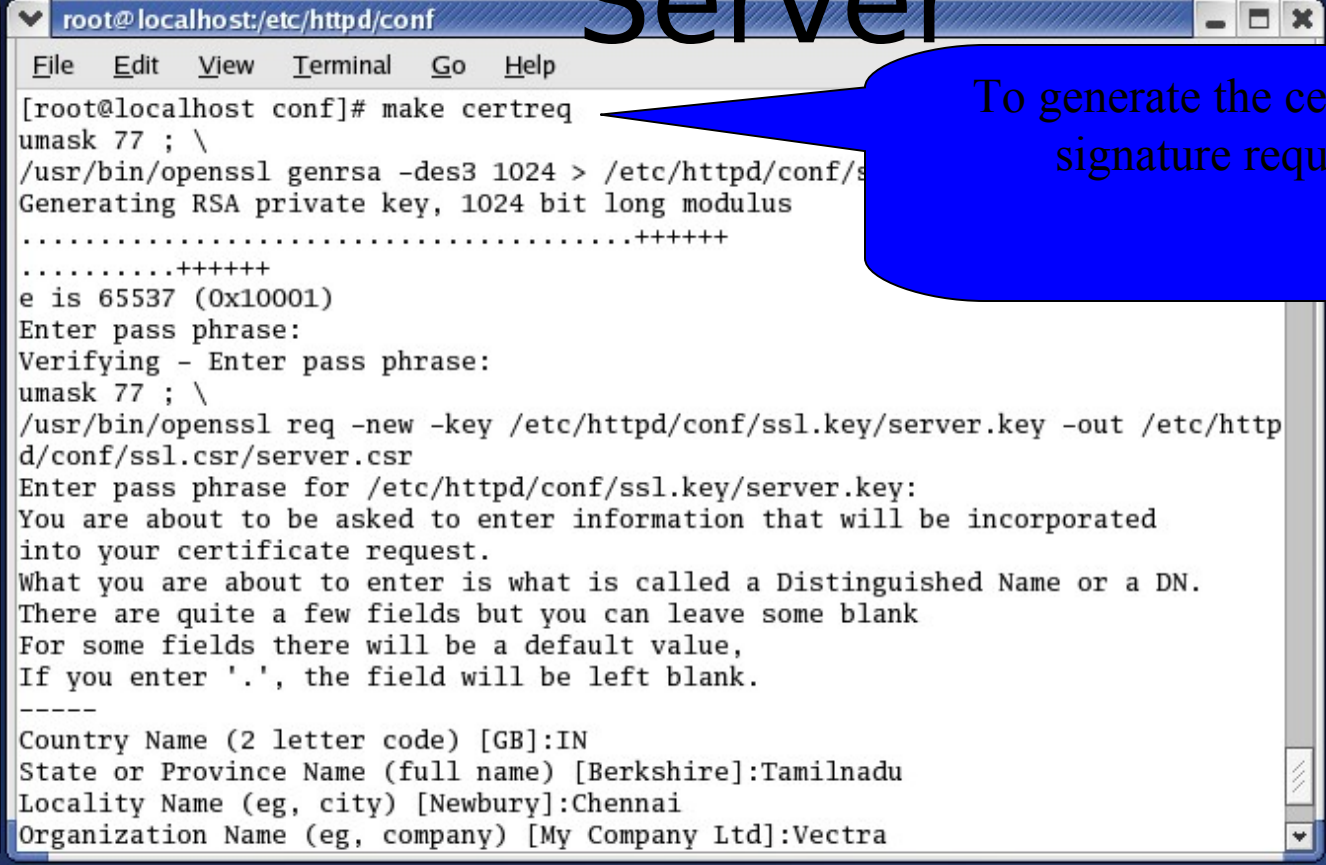


```
root@localhost:/etc/httpd/conf
File Edit View Terminal Go Help
[root@localhost root]# rpm -q mod_ssl
package mod_ssl is not installed
[root@localhost root]# cd /mnt/cdrom/RedHat/RPMS/
[root@localhost RPMS]# rpm -ivh mod_ssl-2.0.46-25.ent.i386.rpm
warning: mod_ssl-2.0.46-25.ent.i386.rpm: V3 DSA signature: NOKEY, key ID db42a60e
Preparing...
1:mod_ssl
[root@localhost RPMS]# cd
[root@localhost root]# cd /etc/httpd/conf
[root@localhost conf]#
```

Query is made to check whether the mod_ssl package is installed. if not, install the packages

Fri Apr 22 5:07 PM

Apache Encrypted Web Server



root@localhost:/etc/httpd/conf

File Edit View Terminal Go Help

```
[root@localhost conf]# make certreq
umask 77 ; \
/usr/bin/openssl genrsa -des3 1024 > /etc/httpd/conf/ssl.key/server.key
Generating RSA private key, 1024 bit long modulus
.....++++++
.....++++++
e is 65537 (0x10001)
Enter pass phrase:
Verifying - Enter pass phrase:
umask 77 ; \
/usr/bin/openssl req -new -key /etc/httpd/conf/ssl.key/server.key -out /etc/httpd/conf/ssl.csr/server.csr
Enter pass phrase for /etc/httpd/conf/ssl.key/server.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [GB]:IN
State or Province Name (full name) [Berkshire]:Tamilnadu
Locality Name (eg, city) [Newbury]:Chennai
Organization Name (eg, company) [My Company Ltd]:Vectra
```

To generate the certificate signature request.

Country Name (2 letter code) [GB]:IN
State or Province Name (full name) [Berkshire]:Tamilnadu
Locality Name (eg, city) [Newbury]:Chennai
Organization Name (eg, company) [My Company Ltd]:Vectra

Fri Apr 22 5:16 PM

Apache Encrypted Web Server



```
root@localhost:/etc/httpd/conf
File Edit View Terminal Go Help
[root@localhost conf]# make testcert
umask 77 ; \
/usr/bin/openssl req -new -key /etc/httpd/conf/ssl.key/server.key -out /etc/httpd/conf/ssl.crt/server.crt
Enter pass phrase for /etc/httpd/conf/ssl.key/server.key:
You are about to be asked to enter information that will be included
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank.
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [GB]:IN
State or Province Name (full name) [Berkshire]:Tamilnadu
Locality Name (eg, city) [Newbury]:Chennai
Organization Name (eg, company) [My Company Ltd]:Vectra
Organizational Unit Name (eg, section) []:Advpro
Common Name (eg, your name or your server's hostname) []:station20
Email Address []:root@station20
[root@localhost conf]#
```

The self signed certificate can be installed using the command make testcert.

Apache Encrypted Web Server



```
root@localhost:/etc/httpd/conf
File Edit View Terminal Go Help
[root@localhost conf]# service httpd restart
Stopping httpd: [ OK ]
Starting httpd: Apache/2.0.46 mod_ssl/2.0.46 (Pass Phrase Dialog)
Some of your private key files are encrypted for security reasons.
In order to read them you have to provide the pass phrases.

Server www.example.com:443 (RSA)
Enter pass phrase:

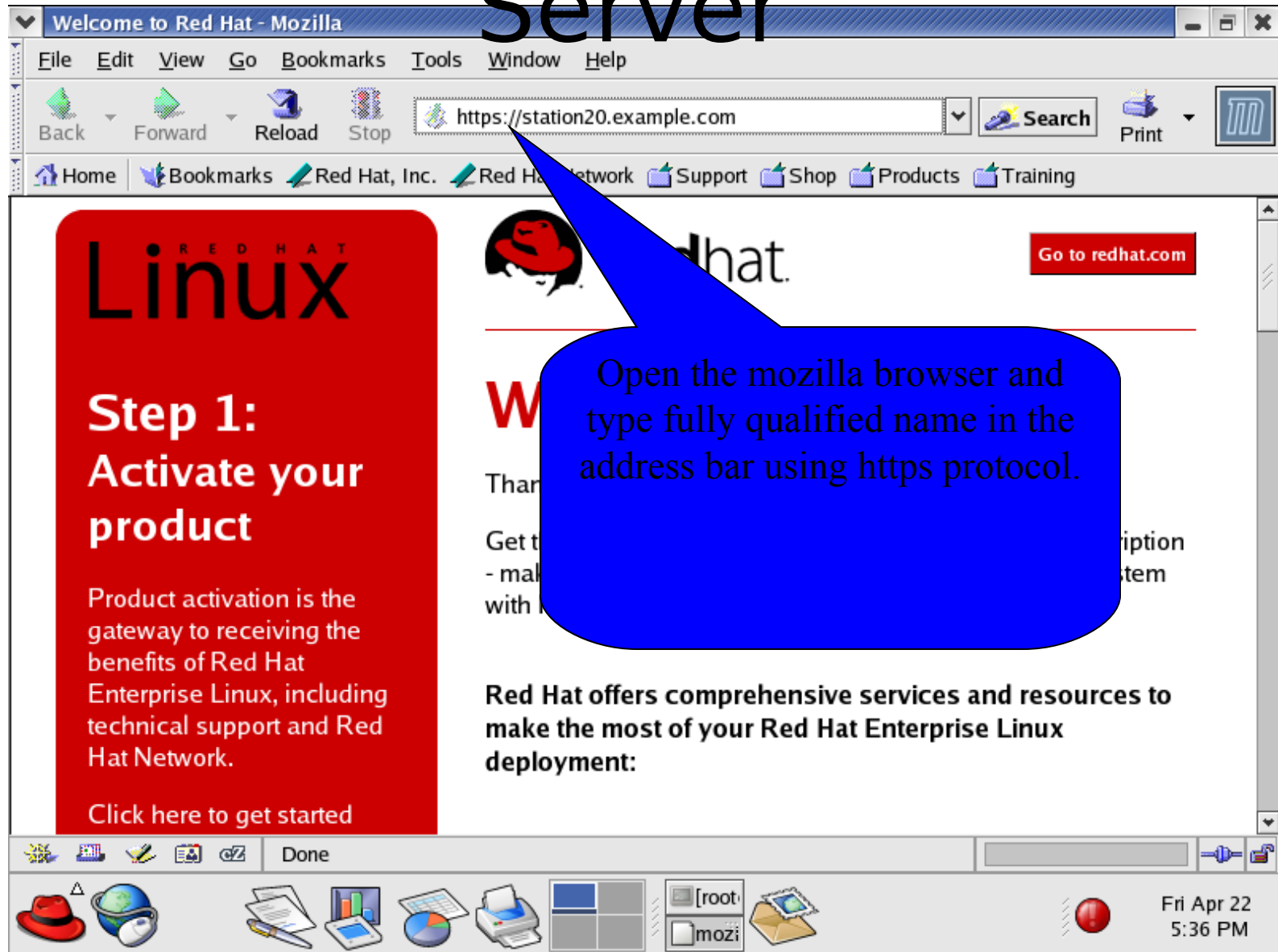
OK: Pass Phrase Dialog successful. [ OK ]
[root@localhost conf]#
```

When we restart the httpd service it will prompts for the password.

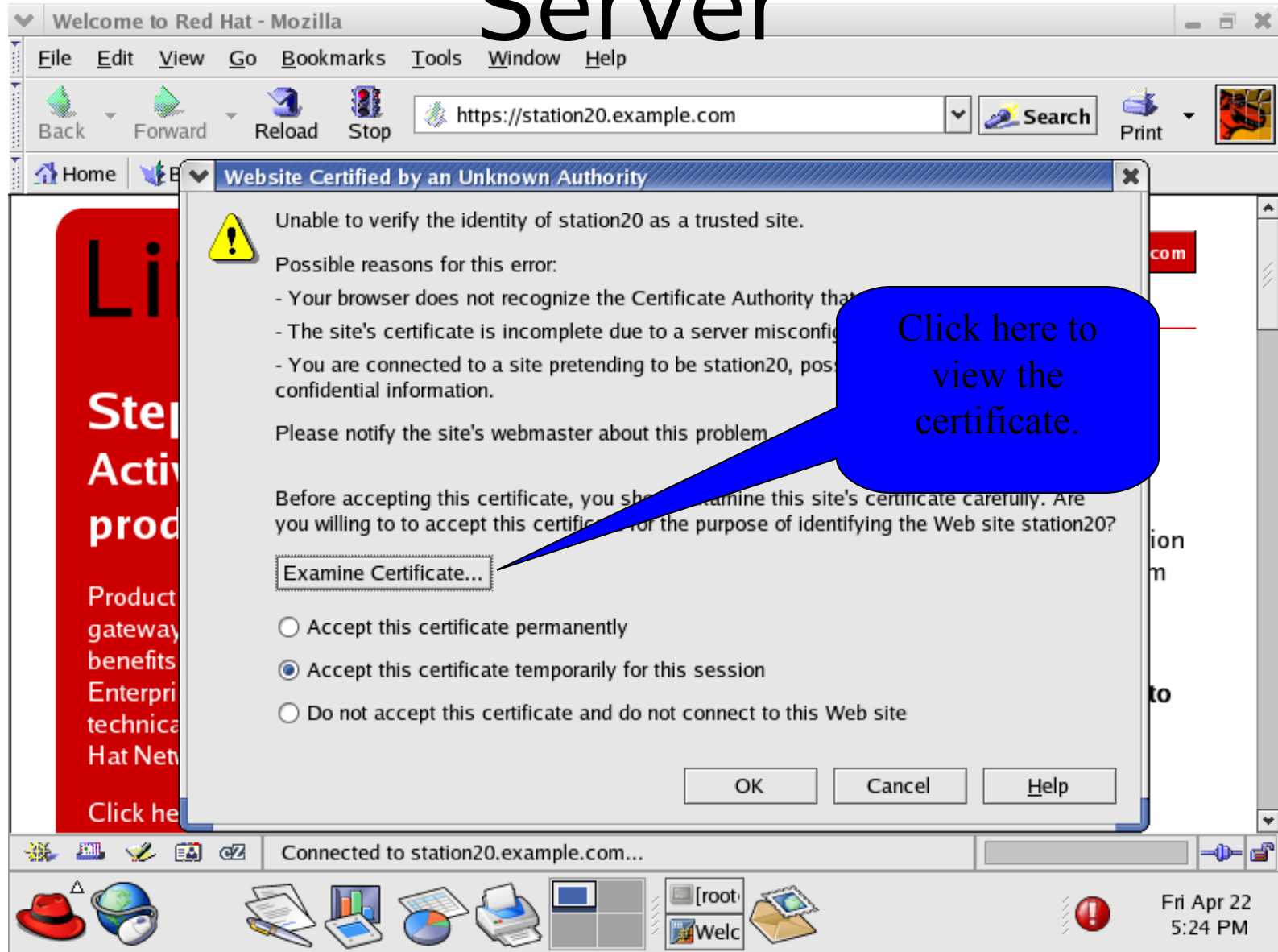


Fri Apr 22
5:19 PM

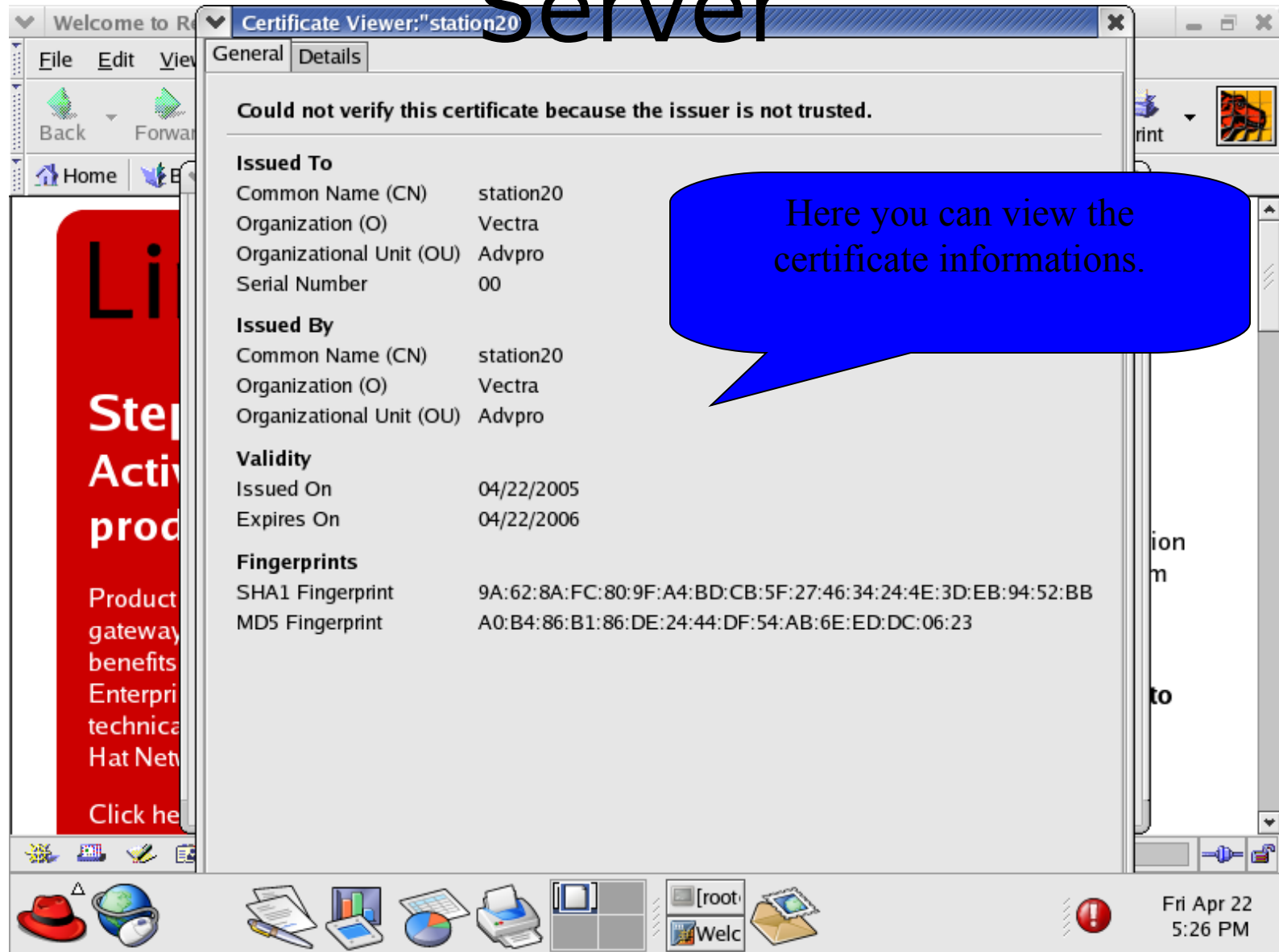
Apache Encrypted Web Server



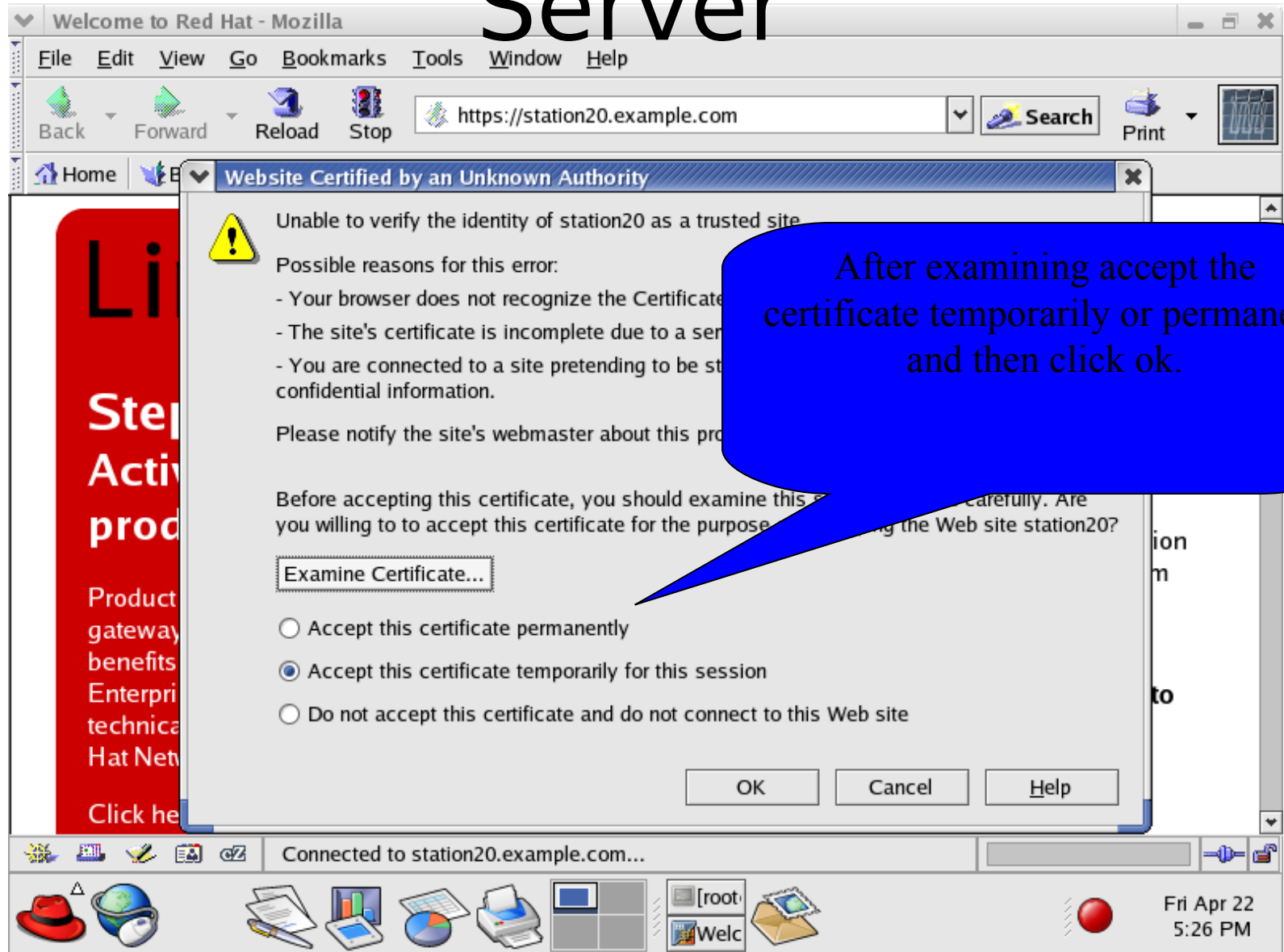
Apache Encrypted Web Server



Apache Encrypted Web Server



Apache Encrypted Web Server



Apache Encrypted Web Server

